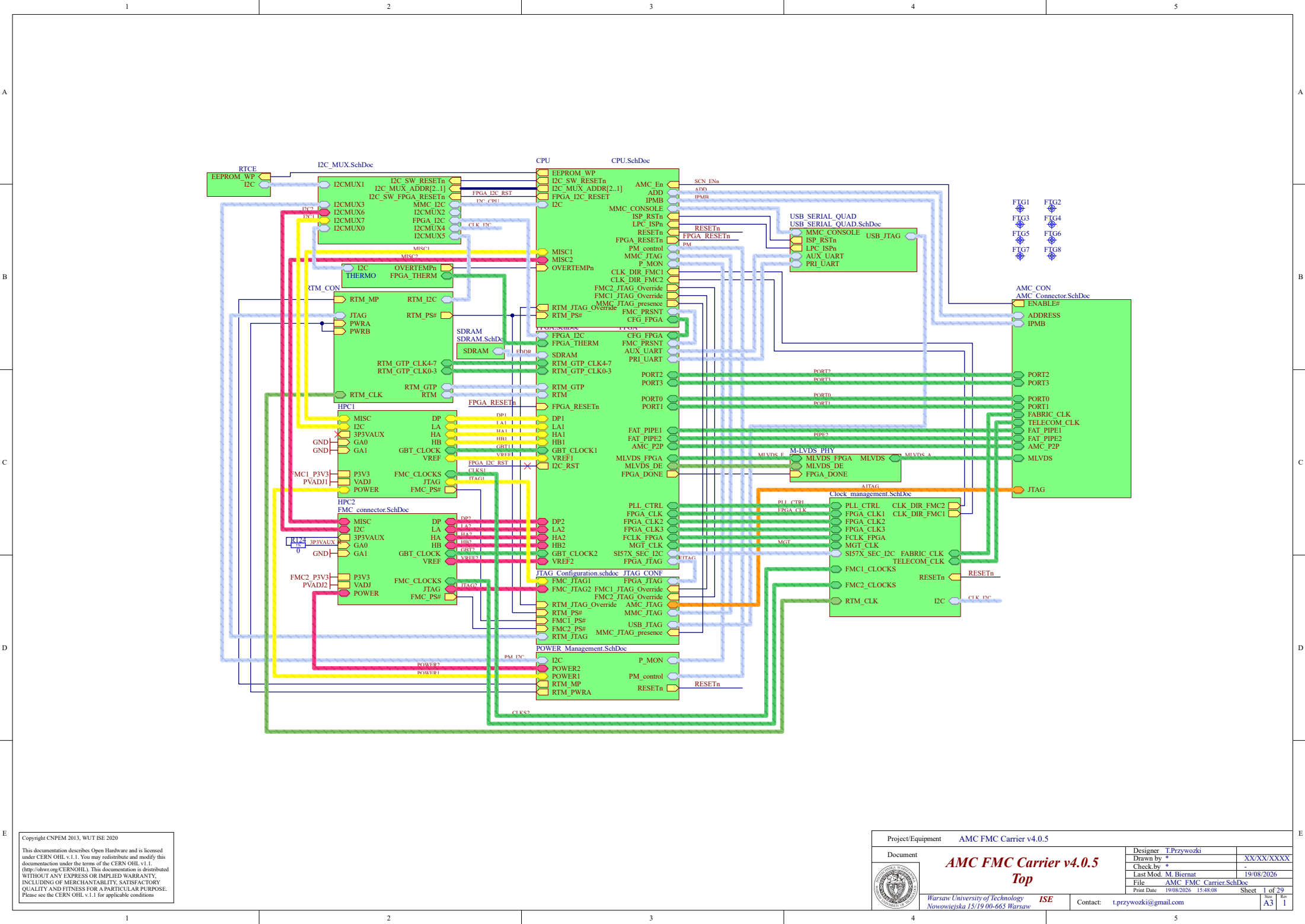




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Project/Equipment		AMC FMC Carrier v4.0.5	
Document		Designer T.Przywozki	
		Drawn by *	XX/XX/XXXX
		Check by *	
		Last Mod. M. Biernat	19/08/2026
		File	AMC FMC Carrier.SchDoc
Print Date		19/08/2026 15:48:08	Sheet 1 of 29
Warsaw University of Technology		ISE	Contact: t.przywozki@gmail.com
Nowowiejska 15/19 00-665 Warsaw			A3 1

A

B

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E

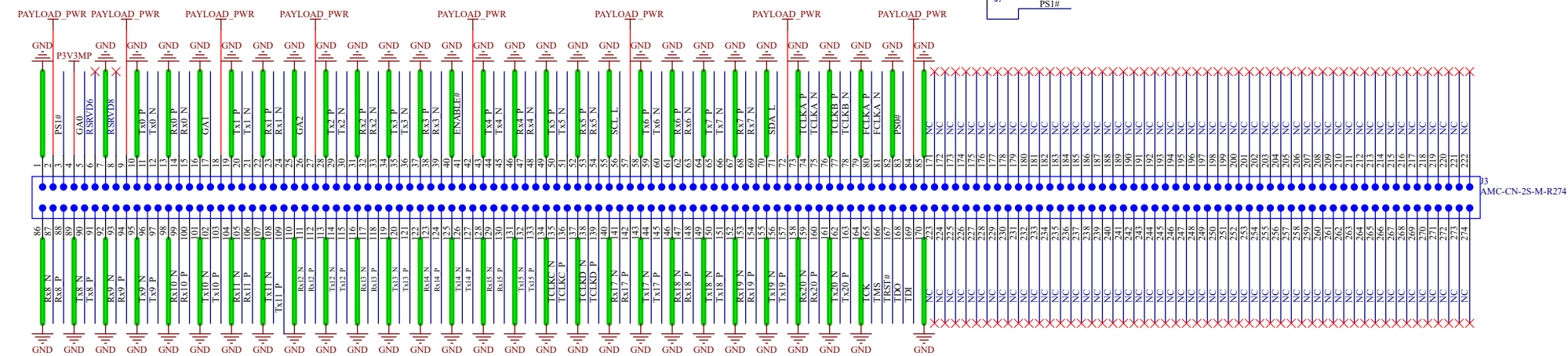
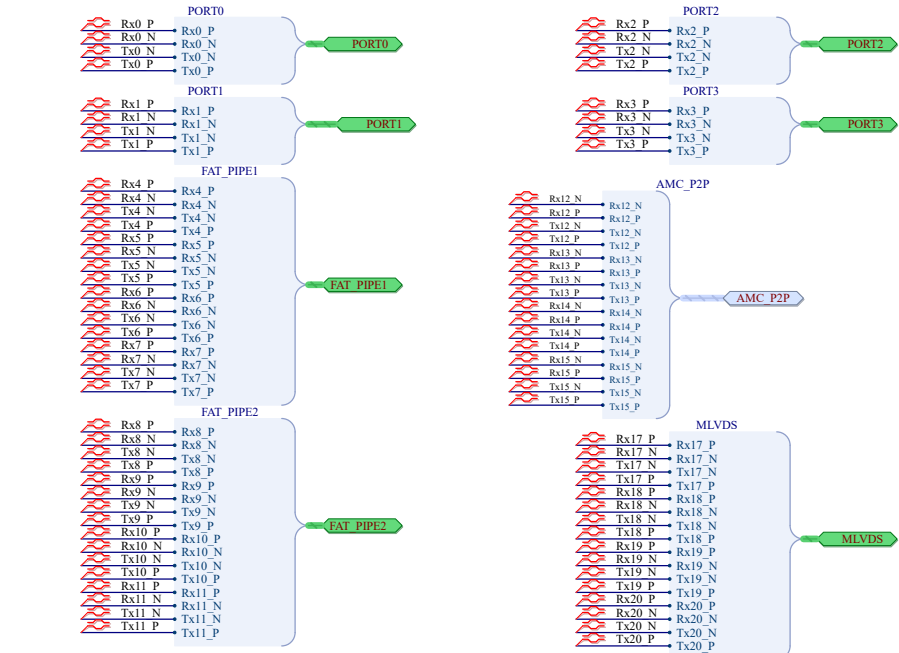
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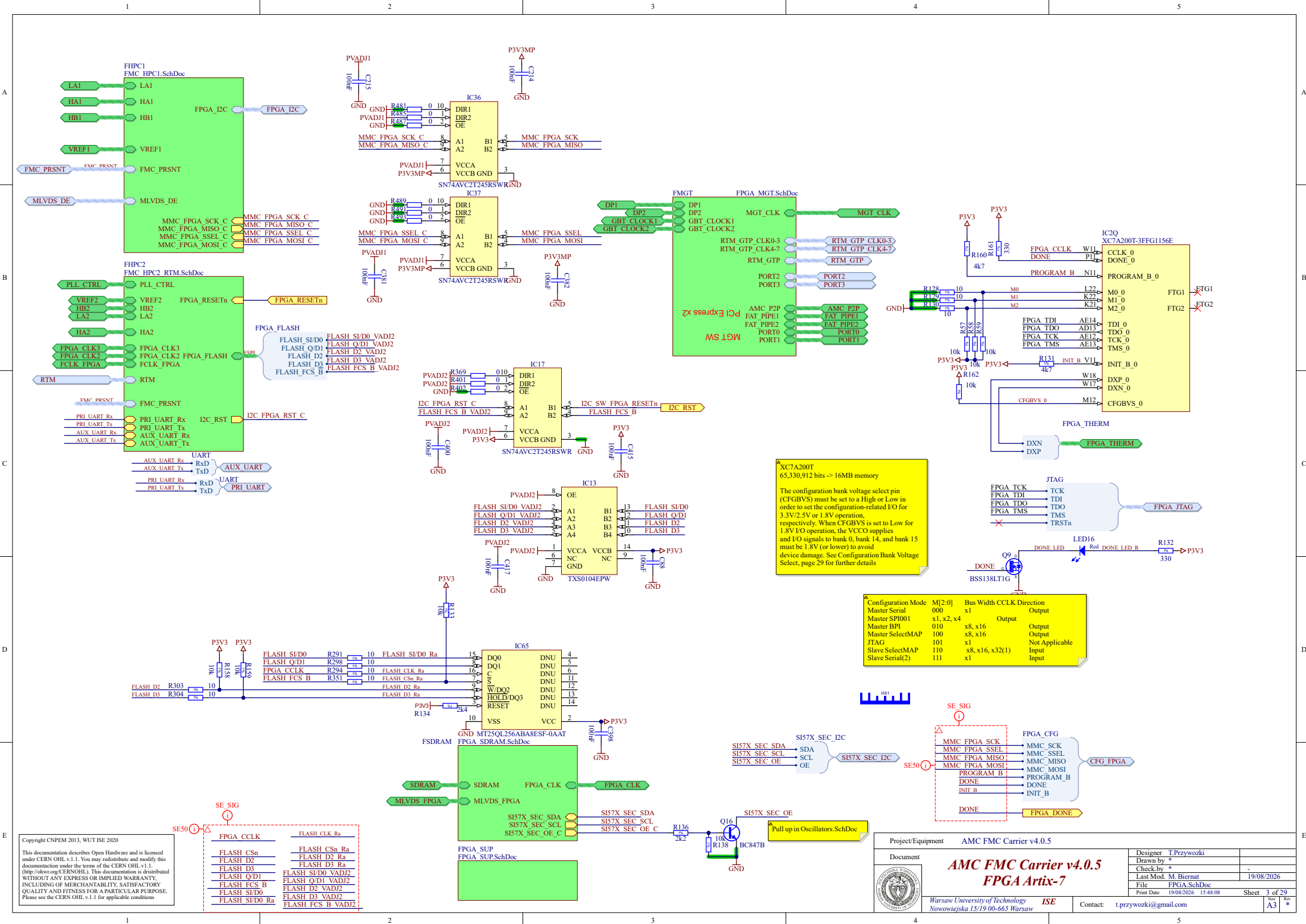
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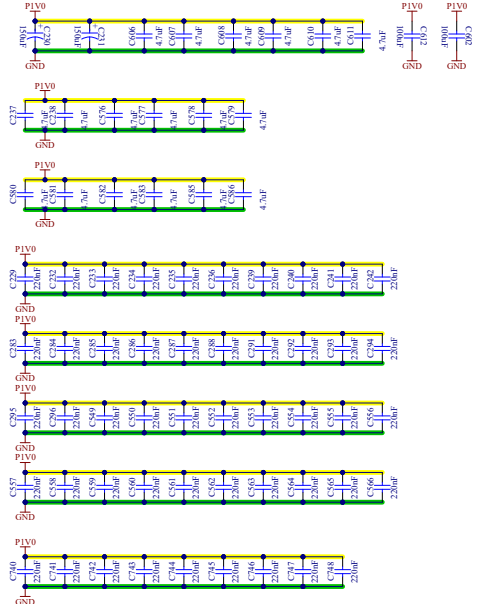
- Dimensions are in MM, nominal values used
- Component height rule derived from AMC Base Specification.PDF, Page 62
- The two corners of outline near the edge-connector are approximated, see AMC Base Specification.PDF, Page 59
- Stackup is not specified in AMC Base Specification.PDF or implemented in this template.

Project/Equipment		AMC FMC Carrier v4.0.5	
Document		AMC FMC Carrier v4.0.5 AMC connector	
	Designer	T.Przywozki	
	Drawn by	-	
	Check by	-	
	Last Mod.	F.Switkowski 19/08/2026	
File		AMC Connector.SchDoc	
Print Date		19/08/2026 15:48:08	Sheet 2 of 29
Contact:		t.przywozki@gmail.com	
Warsaw University of Technology Nowowiejska 15/19 00-665 Warsaw		ISE	
A3		1	



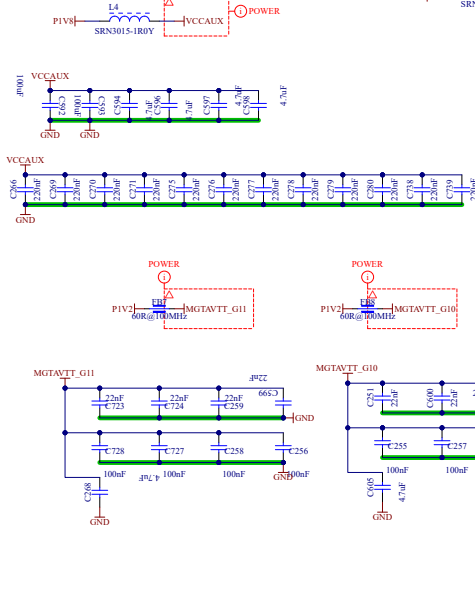
VCCINT

1x680uF, 2x100uF, 28x4.7uF
42x0.47uF on PG.34



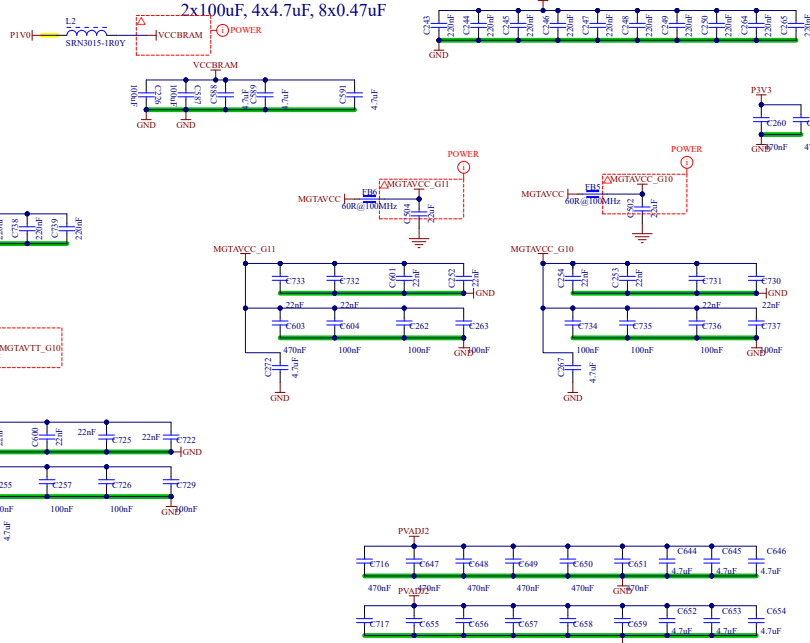
VCCAUX

3x47uF, 4x4.7uF, 7x0.47uF



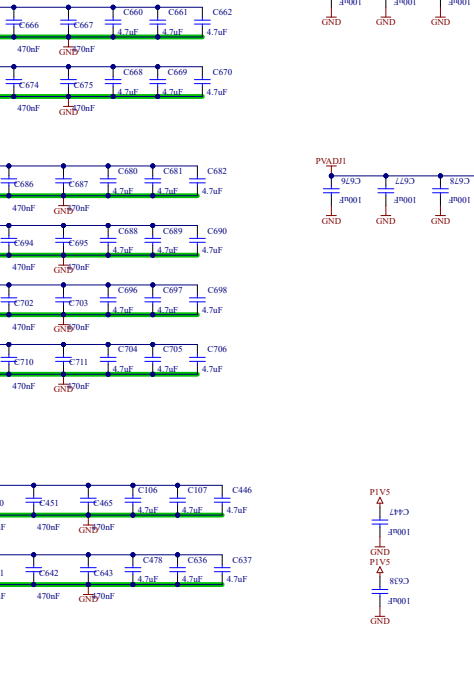
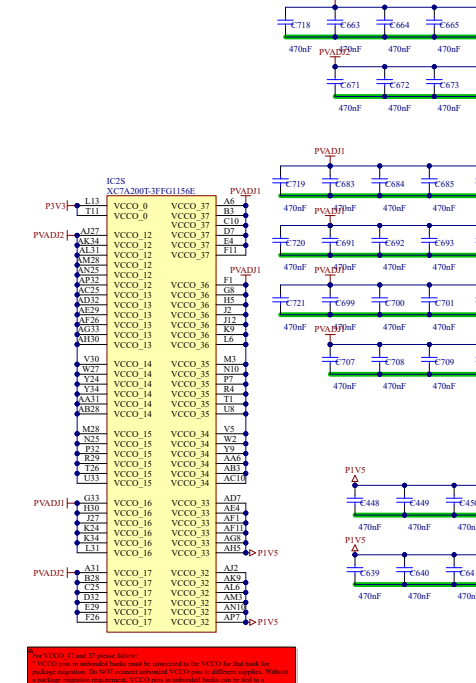
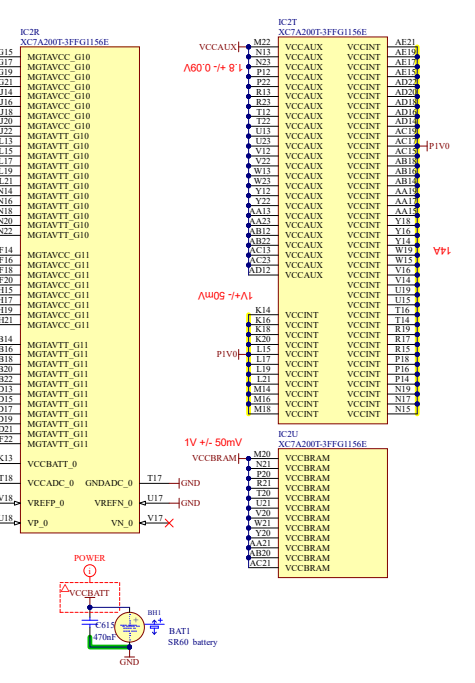
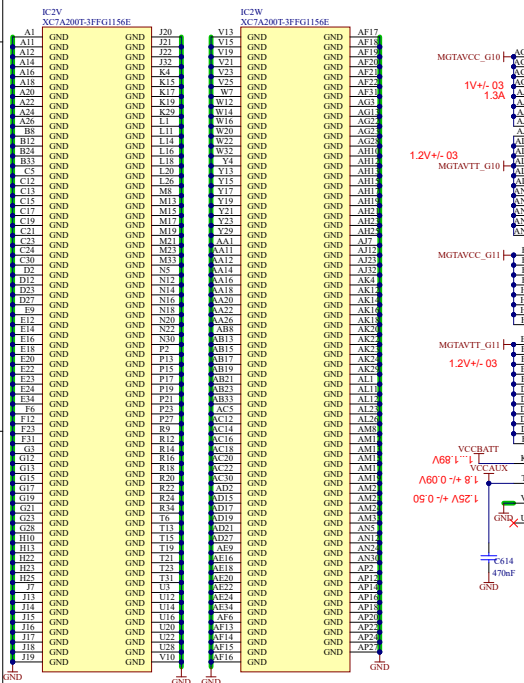
VCCBRAM

2x100uF, 4x4.7uF, 8x0.47uF



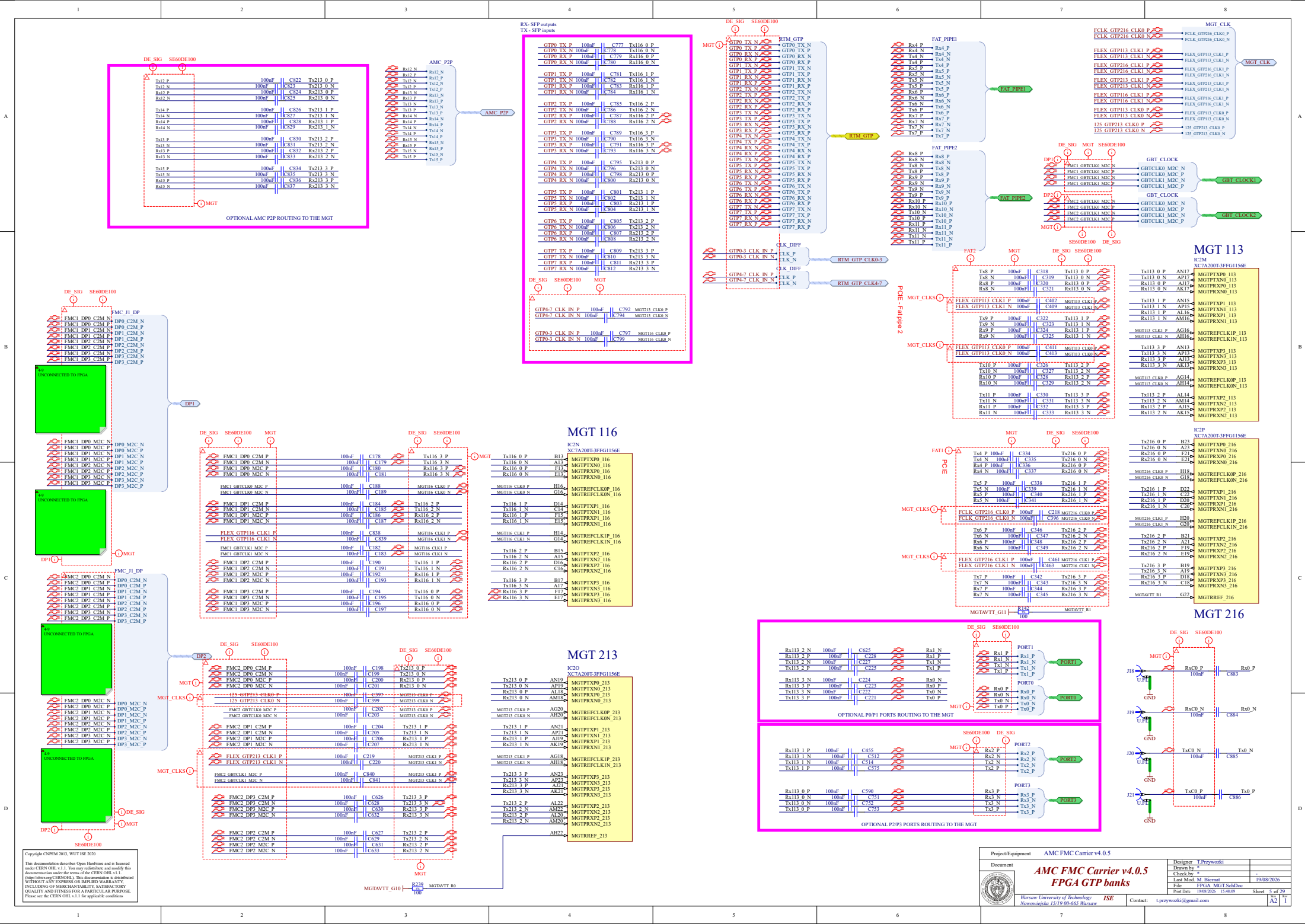
MGTAVCC	1V	2.3A	
MGTAVTT	1.2V	1.5A	
MGTAVCCAUX	1.8V	0.5A	
VCCAUX	1.8V	1A	seq1
VCCAUX_IO	1.8V	0.1A	seq1
VCCBRAM	1V	1.8A	seq1
VCCINT	1V	6A	seq1
VCCO_1.2	1.2V	3.2A	seq2
VCCO_1.35	1.35V	0.5A	seq2
VCCO_1.5	1.5V	0.9A	seq2
VCCO_1.8	1.8V	0.9A	seq2
VCCO_2.5	2.5V	0.9A	seq2
VCCO_3.3	3.3V	0.9A	seq2

power-on sequence is VCCINT, VCCBRAM, VCCAUX, and VCCO
voltage difference between VCCO and VCCAUX must not exceed 2.625V for longer than TVCCO2VCCAUX
power-on sequence to achieve minimum current draw for the GTP transceivers is VCCINT, VMGTAVCC, VMGTAVTT or VMGTAVCCAUX, VCCINT, VMGTAVTT. There is no recommended sequencing for VMGTAVCCAUX. Both VMGTAVCC and VCCINT can be ramped simultaneously. The recommended power-off sequence is the reverse of the power-on sequence to achieve minimum current draw.
When VMGTAVTT is powered before VMGTAVCC and VMGTAVTT - VMGTAVCC > 150 mV and VMGTAVCC < 0.7V, the VMGTAVTT current draw can increase by 460 mA per transceiver during VMGTAVCC ramp up. The duration of the current draw can be up to 0.3 x VMGTAVCC (ramp time from GND to 90% of VMGTAVCC). The reverse is true for power-down.
When VMGTAVTT is powered before VCCINT and VMGTAVTT - VCCINT > 150 mV and VCCINT < 0.7V, the VMGTAVTT current draw can increase by 50 mA per transceiver during VCCINT ramp up. The duration of the current draw can be up to 0.3 x TVCCINT (ramp time from GND to 90% of VCCINT). The reverse is true for power-down.



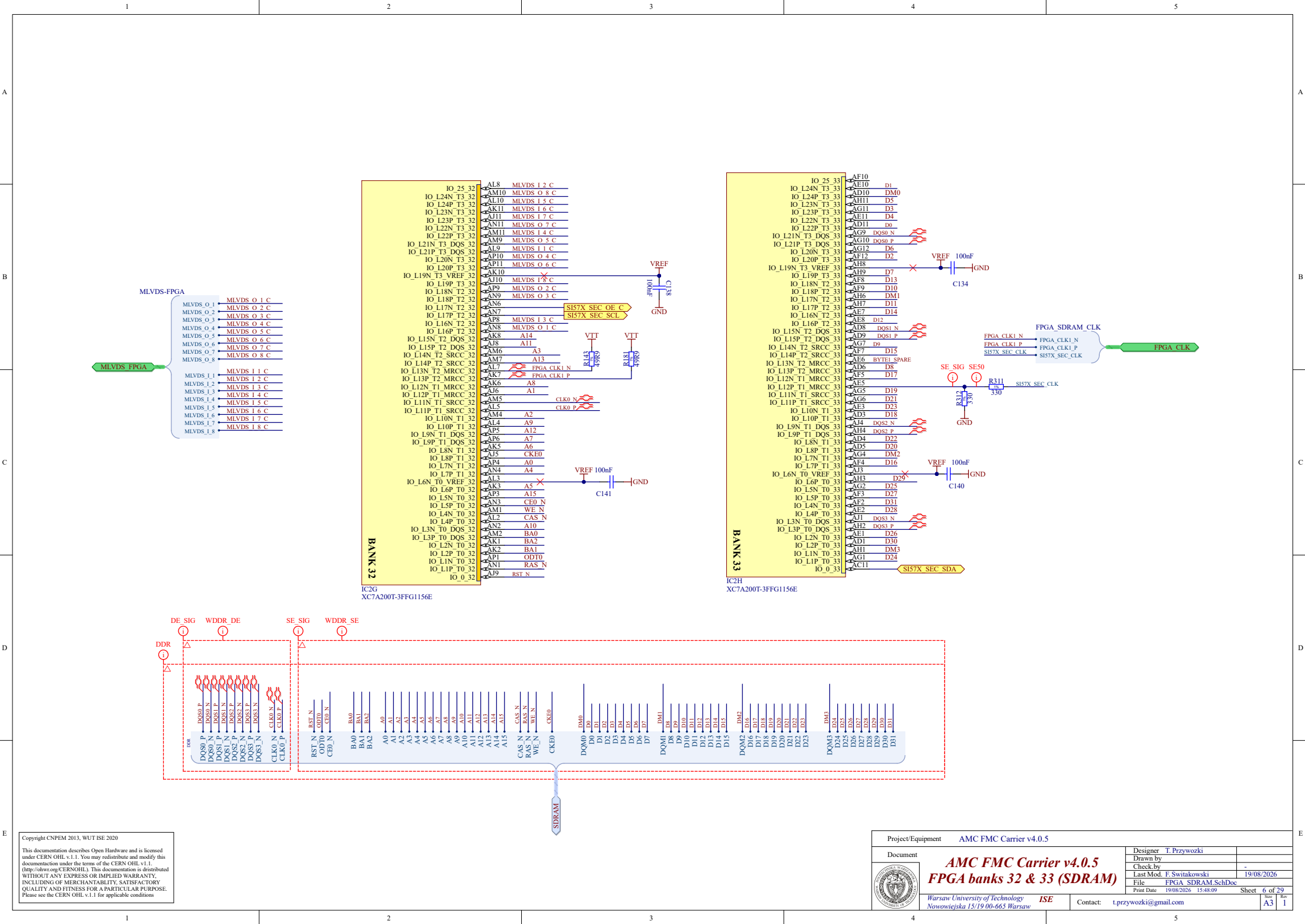
Two VCCO_1.7 and 0.9 plane below
VCCO pin is connected to both and is connected to the VCCO pin for both the package designated. Do NOT connect individual VCCO pins to different supplies. Without package designator, VCCO pin is individual both pins to be fed a common supply (VCCO) for general.
No magnetic materials closer than 600 mils (1.524 mm) to the VCCO pin.
and C100

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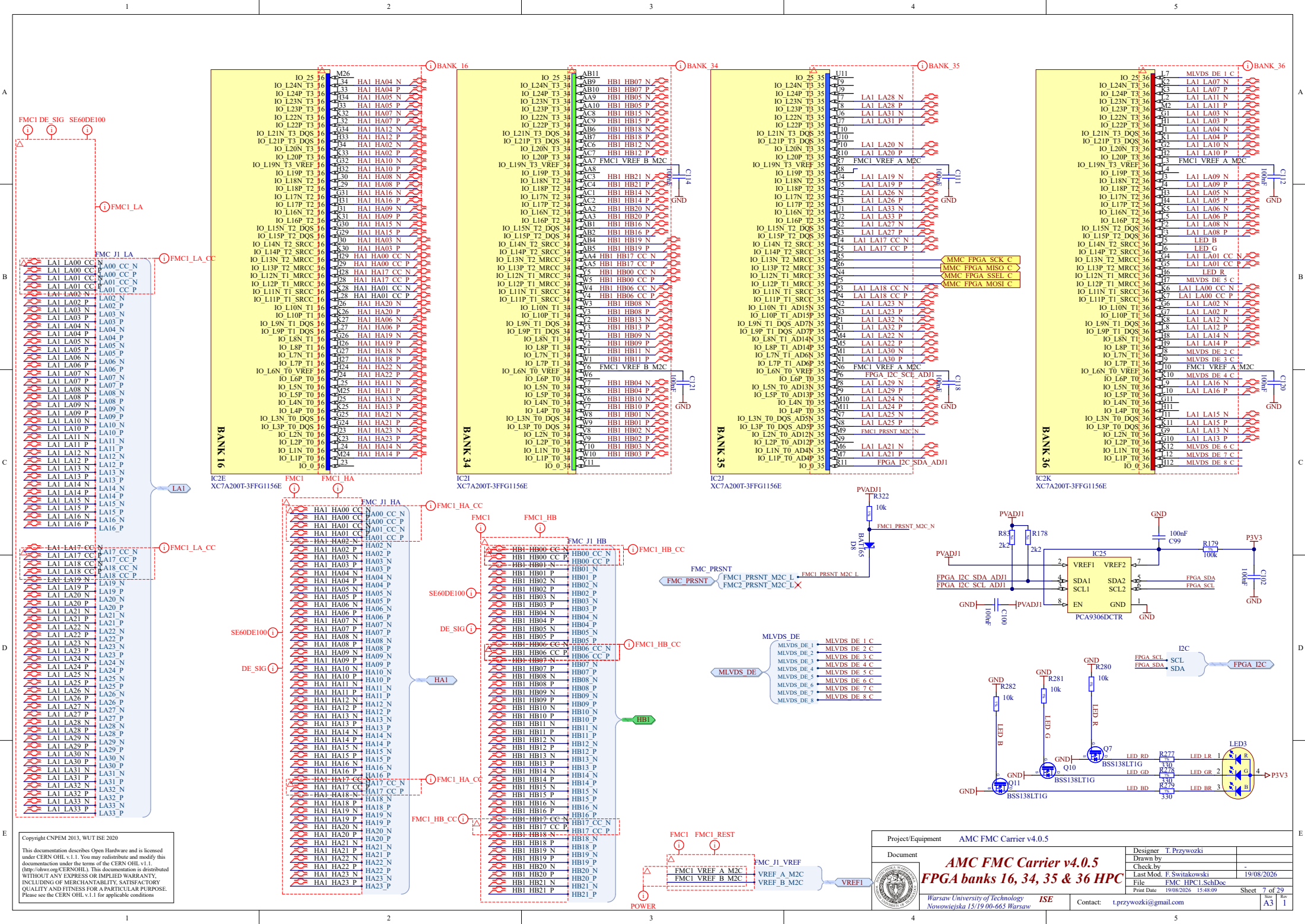
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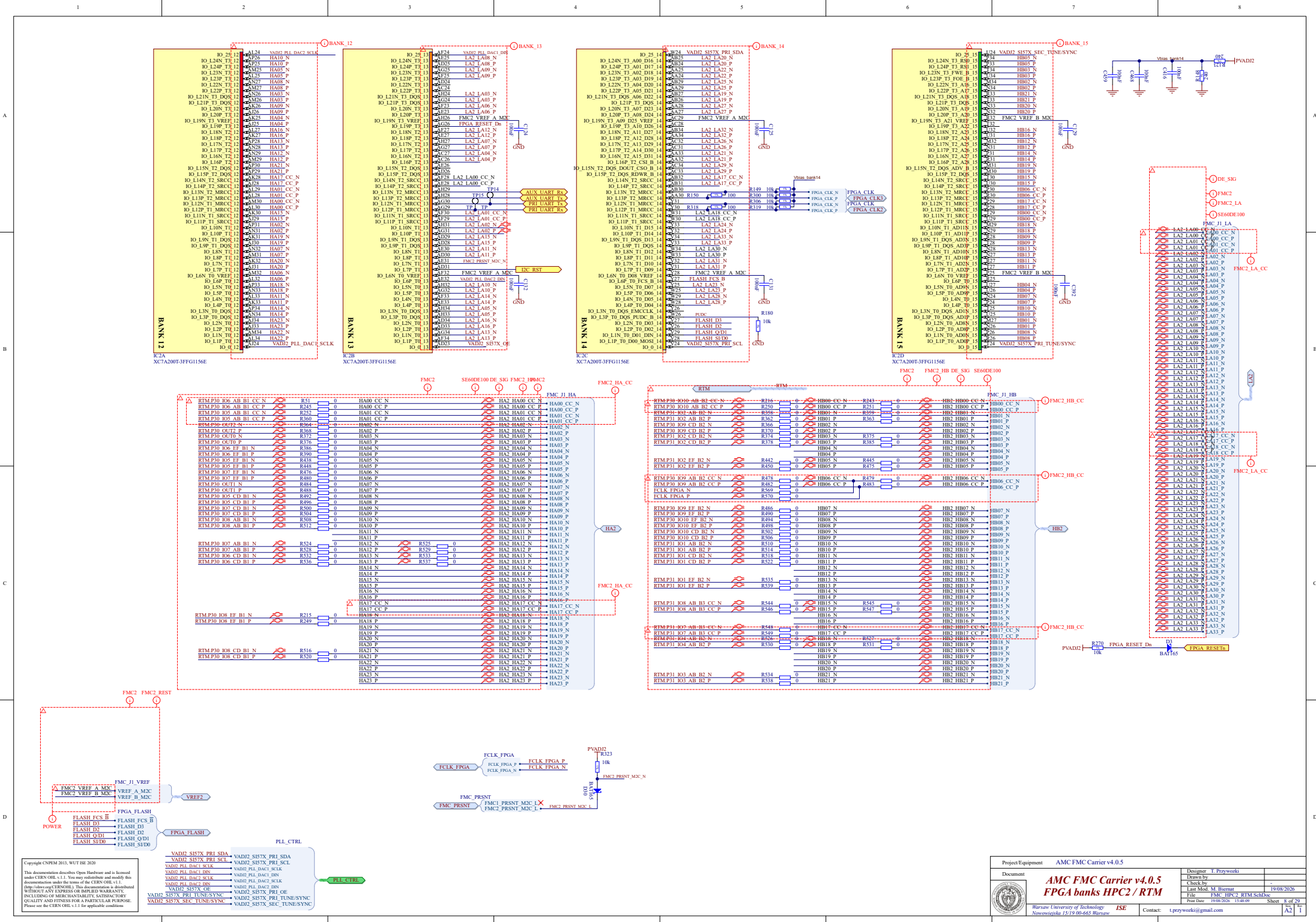


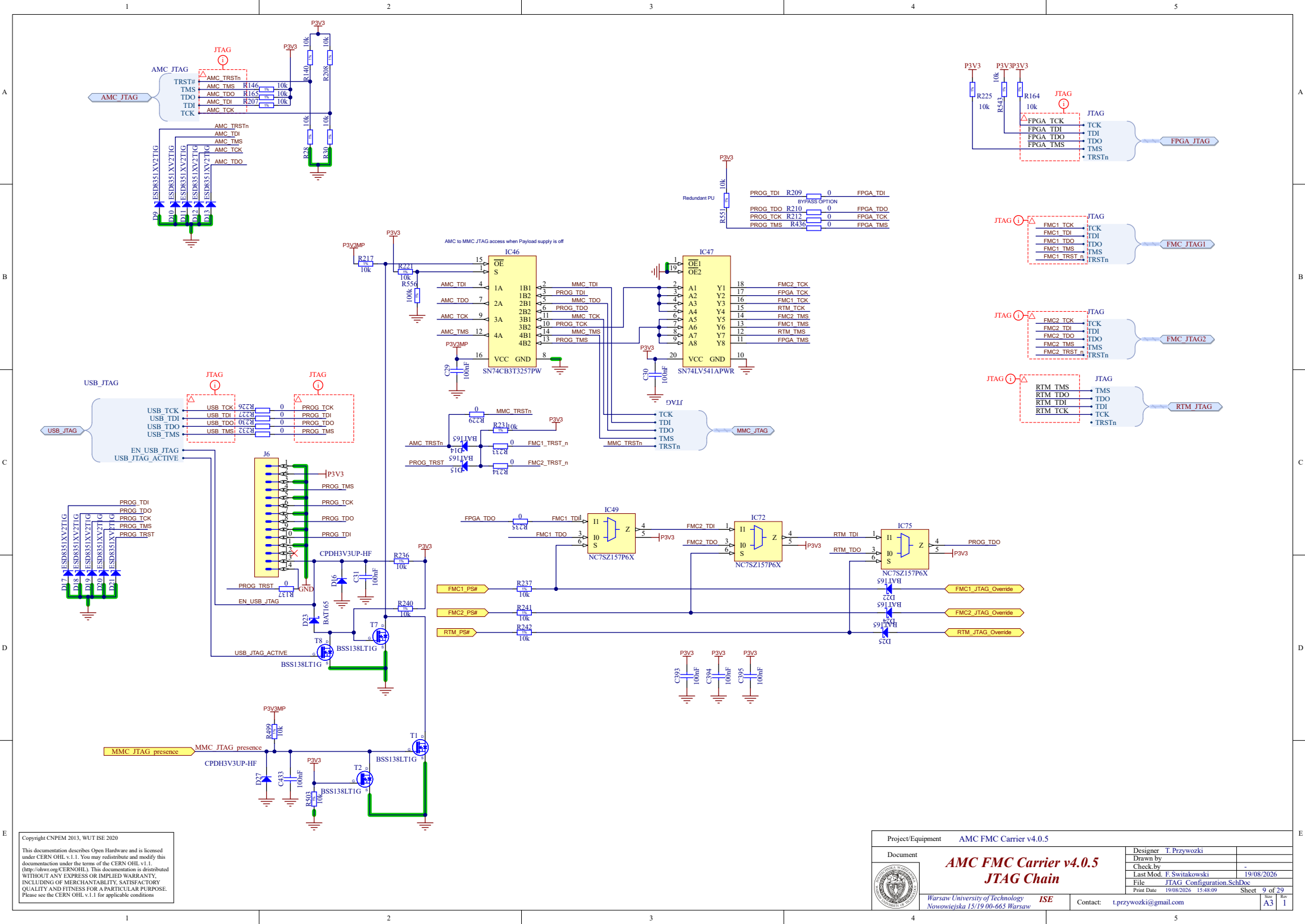
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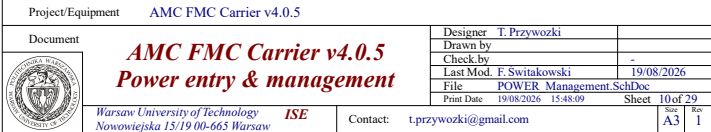
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		Last Mod. F. Switkowski 19/08/2026	
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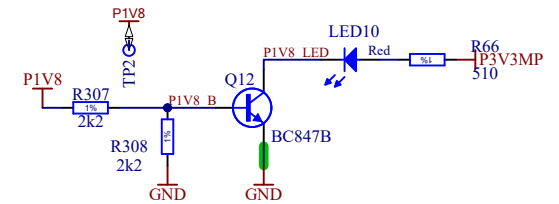
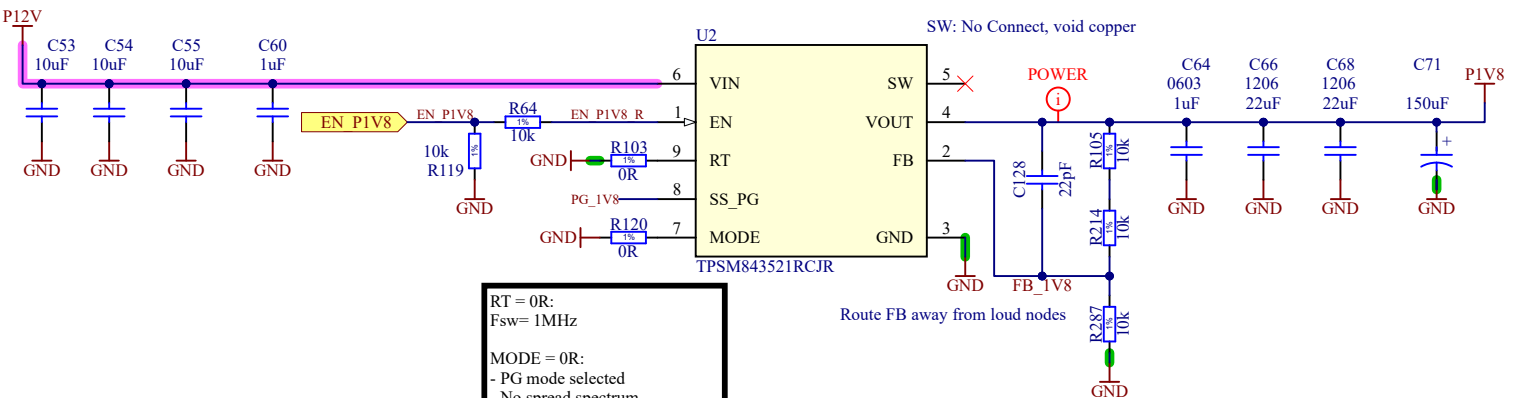
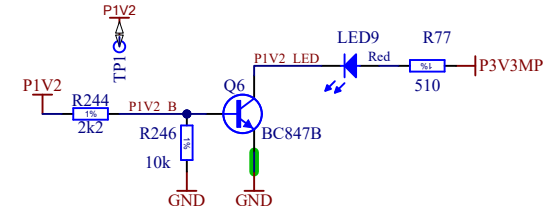
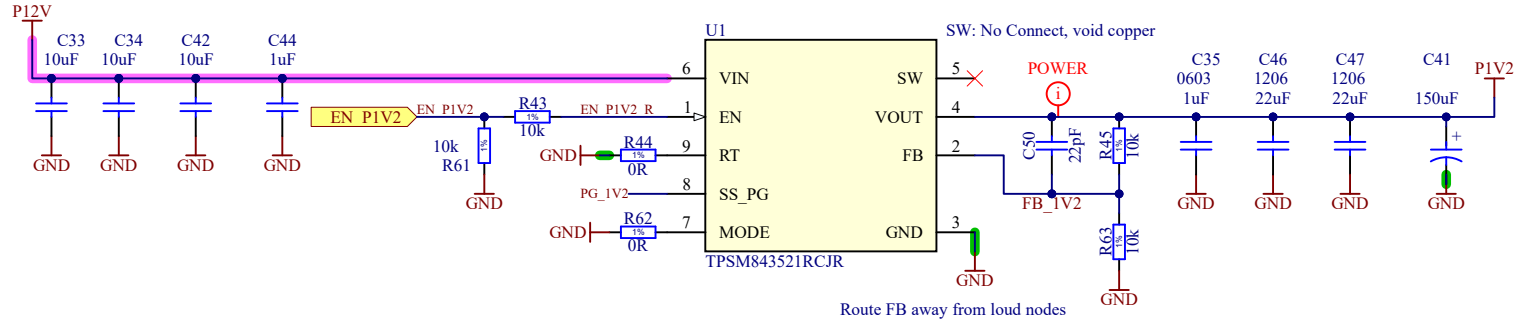
8







Compatible parts:
TPSM843521 - 5A
TPSM843321 - 3A
TPSM84538 - 5A, 28V in
TPSM84338 - 3A, 28V in

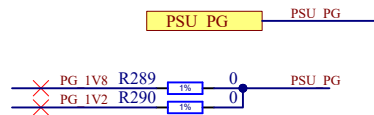


RT = 0R:
Fsw = 1MHz

MODE = 0R:
- PG mode selected
- No spread spectrum
- FCCS mode

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Project/Equipment AMC FMC Carrier v4.0.5

Document



AMC FMC Carrier v4.0.5 Power P1V2 & P1V8

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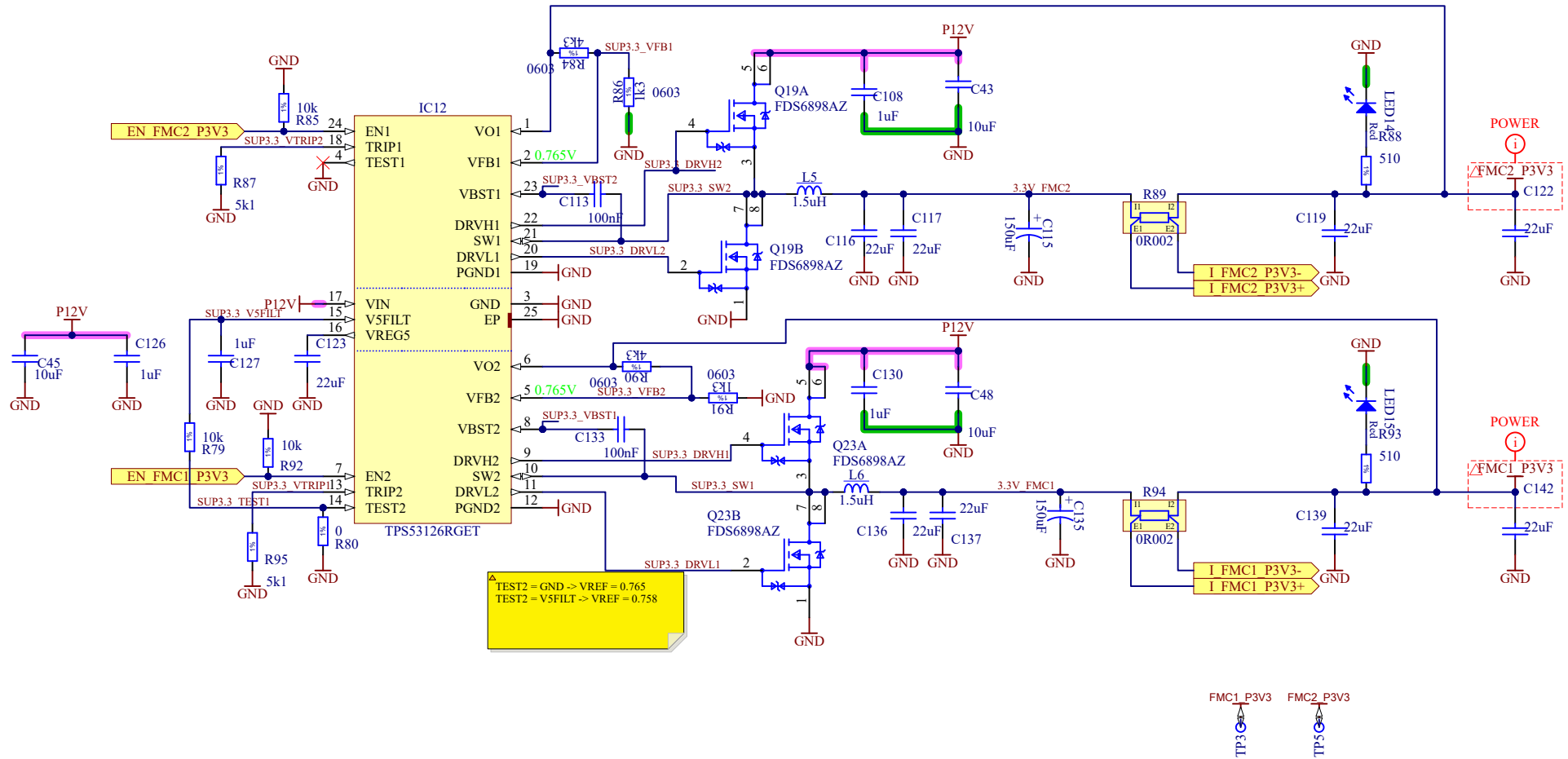
ISE

Contact: t.przywozki@gmail.com

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Drawn by		XX/XX/XXXX
Check by		-
Last Mod.	F. Świątkowski	19/08/2026
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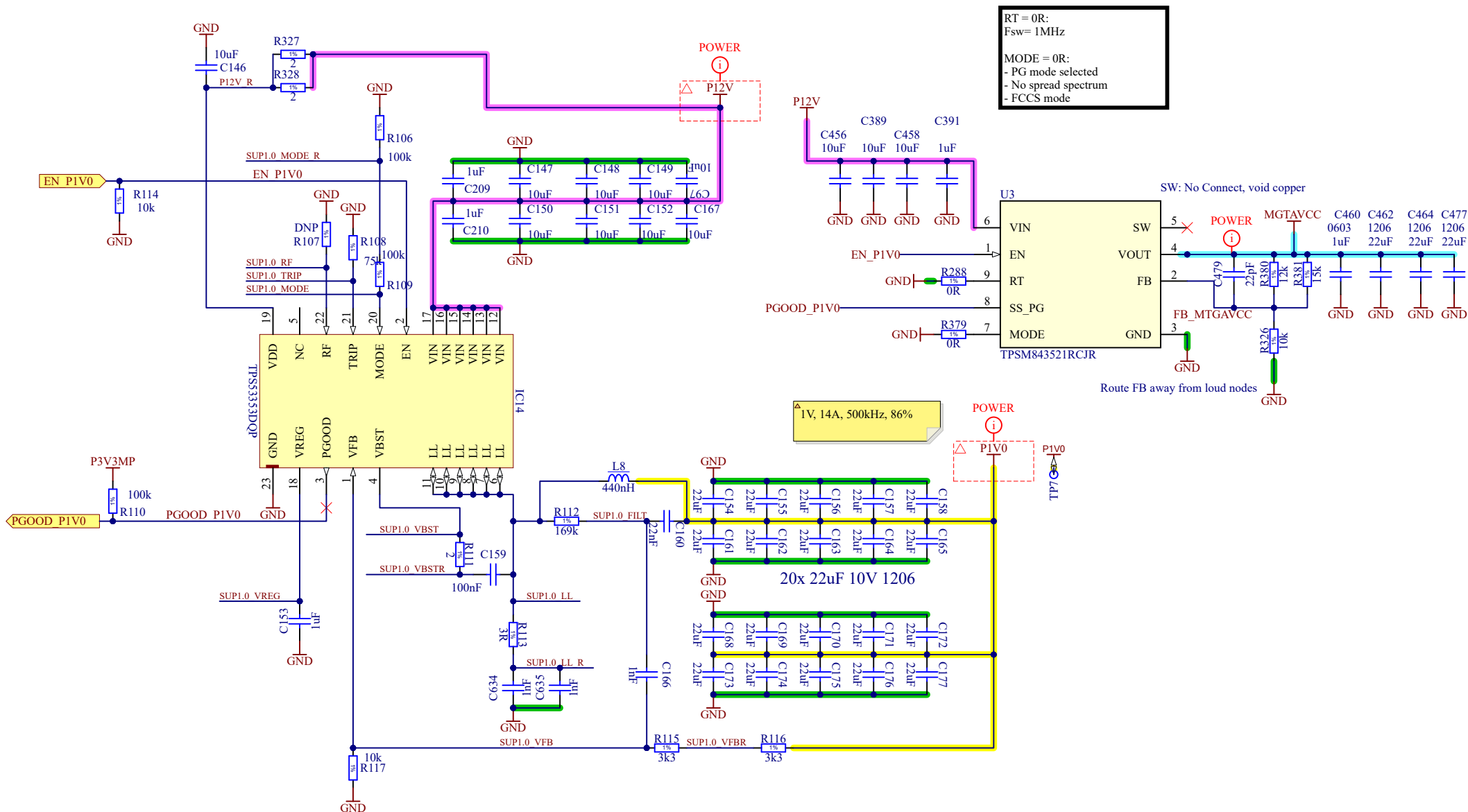
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Warsaw University of Technology Nowowiejska 15/19 00-665 Warsaw		ISE	Contact: t.przywozki@gmail.com
Print Date 19/08/2026 15:48:10		Sheet 12 of 29	Size A4 Rev 1



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Project/Equipment AMC FMC Carrier v4.0.5

Document

AMC FMC Carrier v4.0.5 Power P1V0



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Designer T. Przywozki

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Last Mod. F. Świątkowski

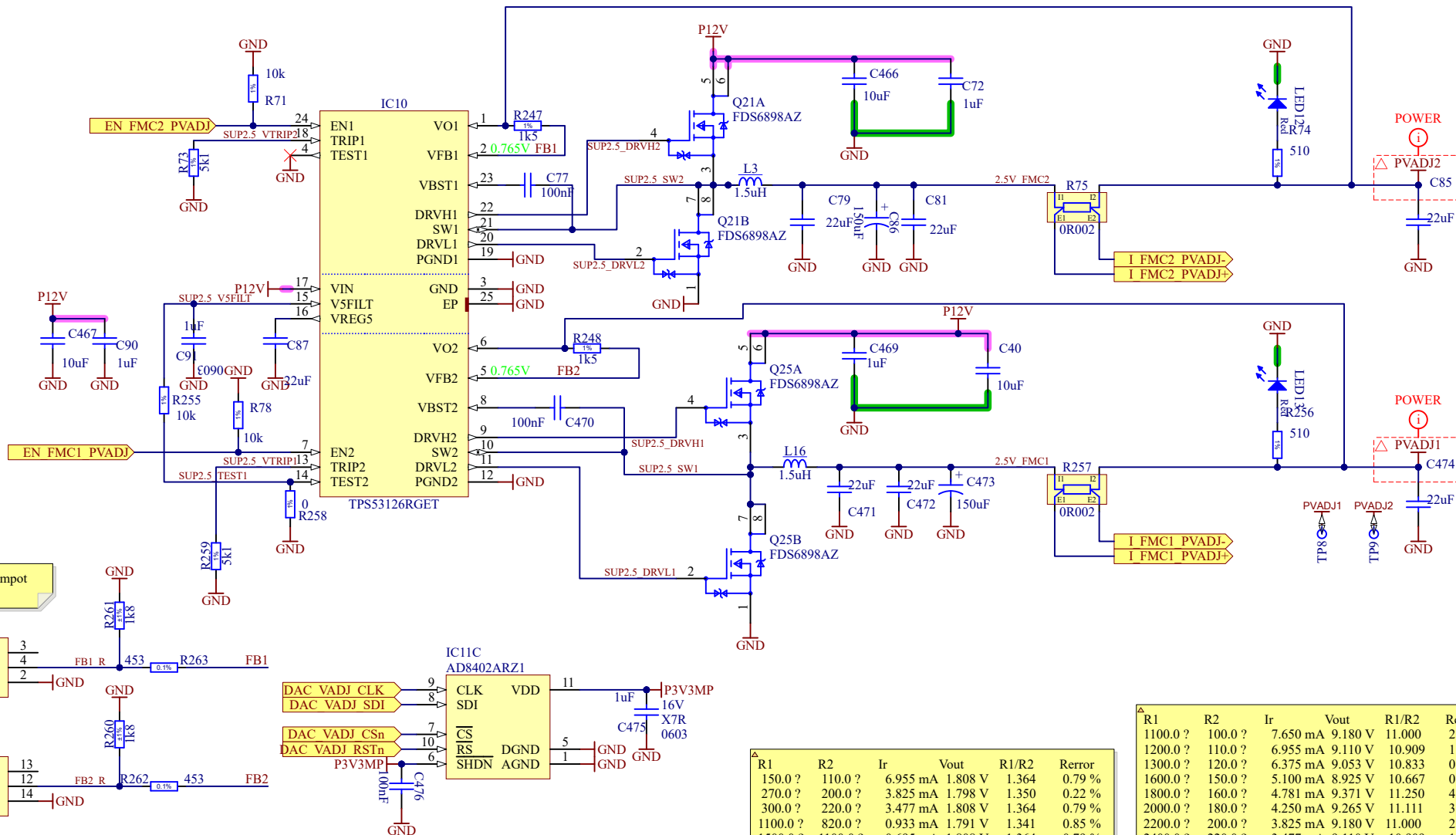
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Project/Equipment AMC FMC Carrier v4.0.5

Document



AMC FMC Carrier v4.0.5 Power FMCs PVADJ

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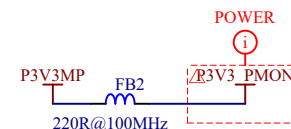
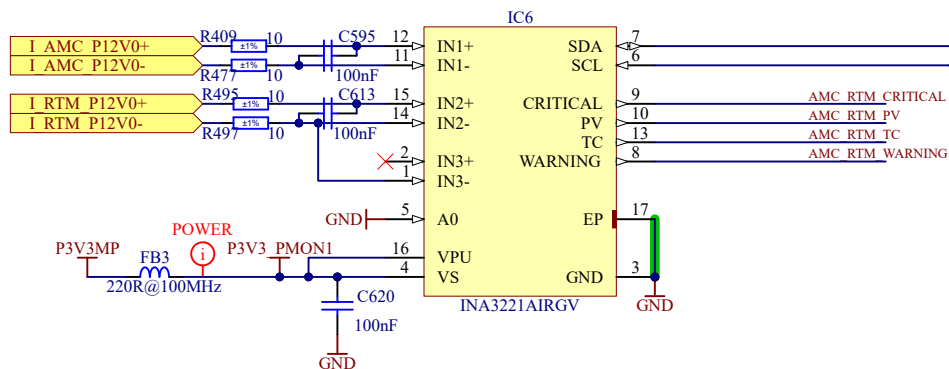
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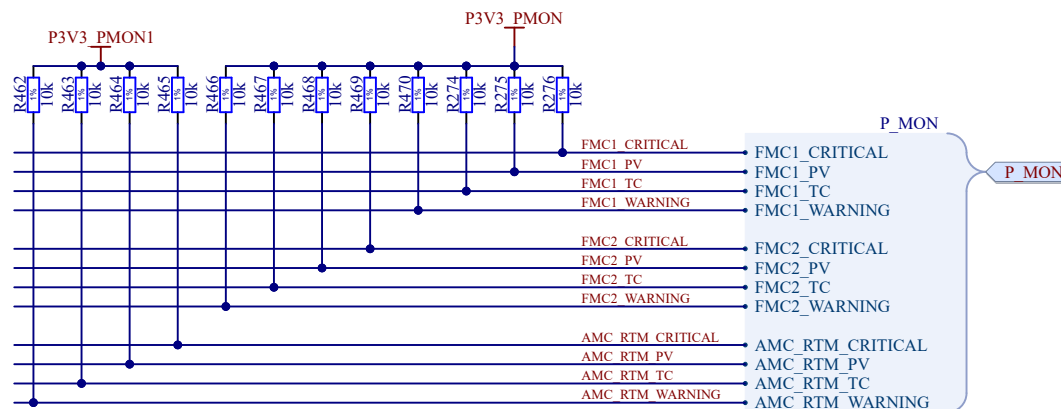
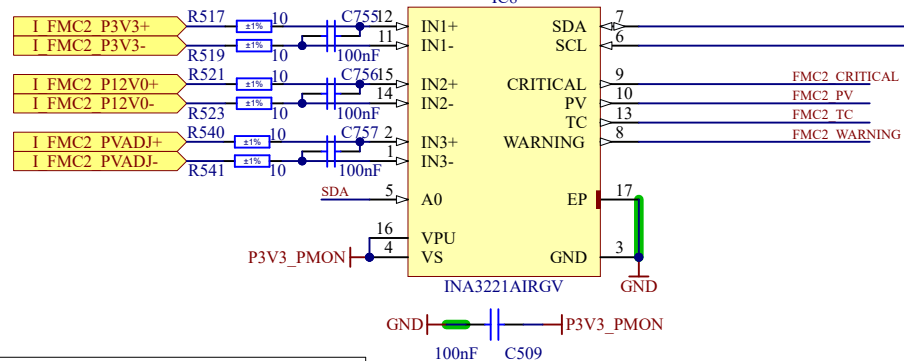
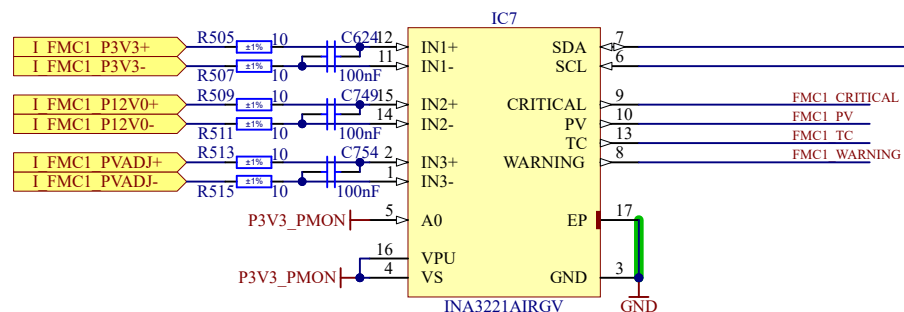
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Rev 1

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1200.0 ?	110.0 ?	6.955 mA	9.110 V	10.909	1.34 %
1300.0 ?	120.0 ?	6.375 mA	9.053 V	10.833	0.64 %
1600.0 ?	150.0 ?	5.100 mA	8.925 V	10.667	0.91 %
1800.0 ?	160.0 ?	4.781 mA	9.371 V	11.250	4.51 %
2000.0 ?	180.0 ?	4.250 mA	9.265 V	11.111	3.22 %
2200.0 ?	200.0 ?	3.825 mA	9.180 V	11.000	2.19 %
2400.0 ?	220.0 ?	3.477 mA	9.110 V	10.909	1.34 %
2700.0 ?	240.0 ?	3.188 mA	9.371 V	11.250	4.51 %
3000.0 ?	270.0 ?	2.833 mA	9.265 V	11.111	3.22 %
3300.0 ?	300.0 ?	2.550 mA	9.180 V	11.000	2.19 %



A1	SLAVE ADDRESS
GND	1000000
VS	1000001
SDA	1000010
SCL	1000011



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Project/Equipment AMC FMC Carrier v4.0.5

Document

AMC FMC Carrier v4.0.5 Power Measurement



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Contact: t.przywozki@gmail.com

Designer T. Przywozki

Drawn by

XX/XX/XXXX

Check by

-

Last Mod. F. Świątkowski

19/08/2026

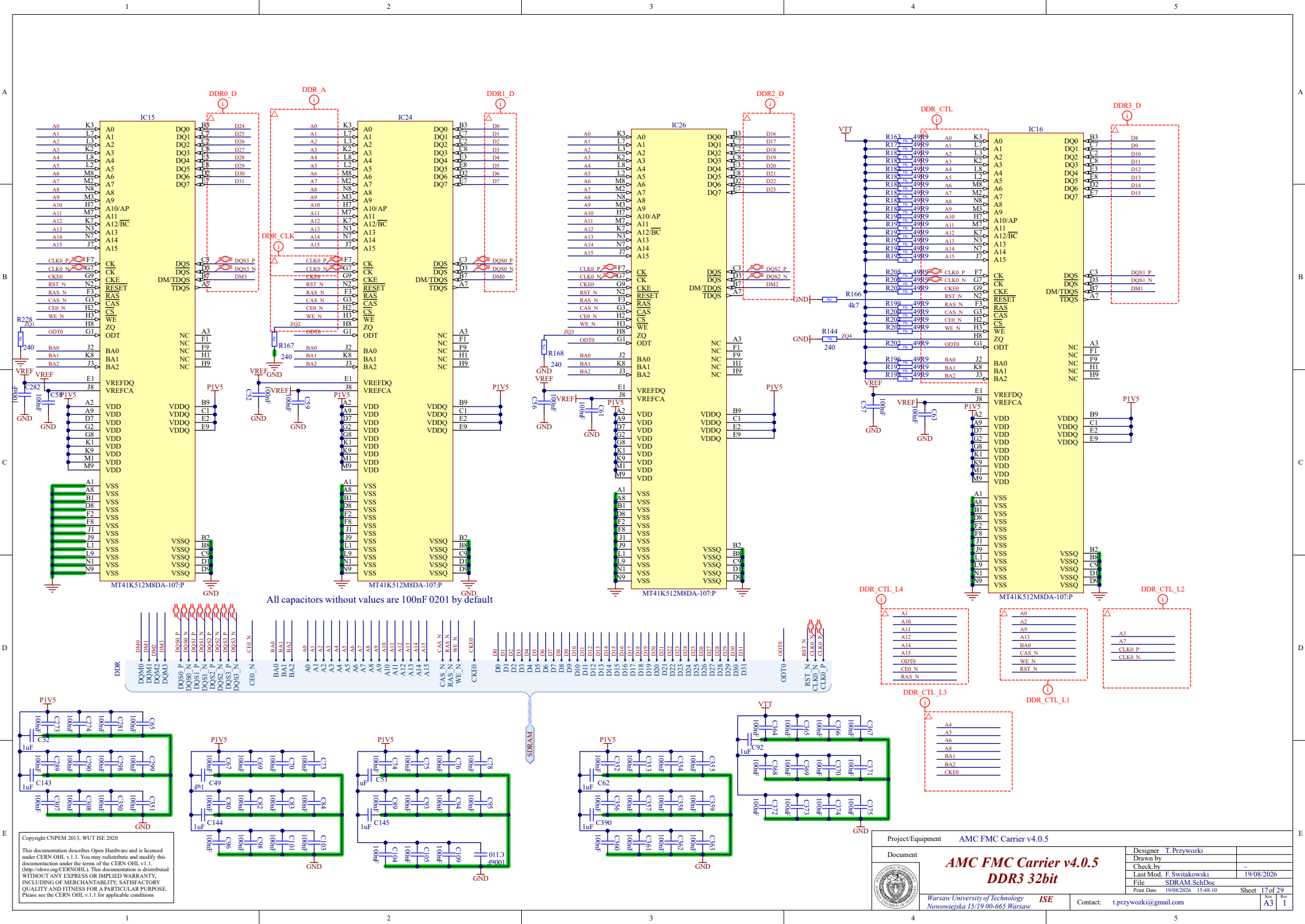
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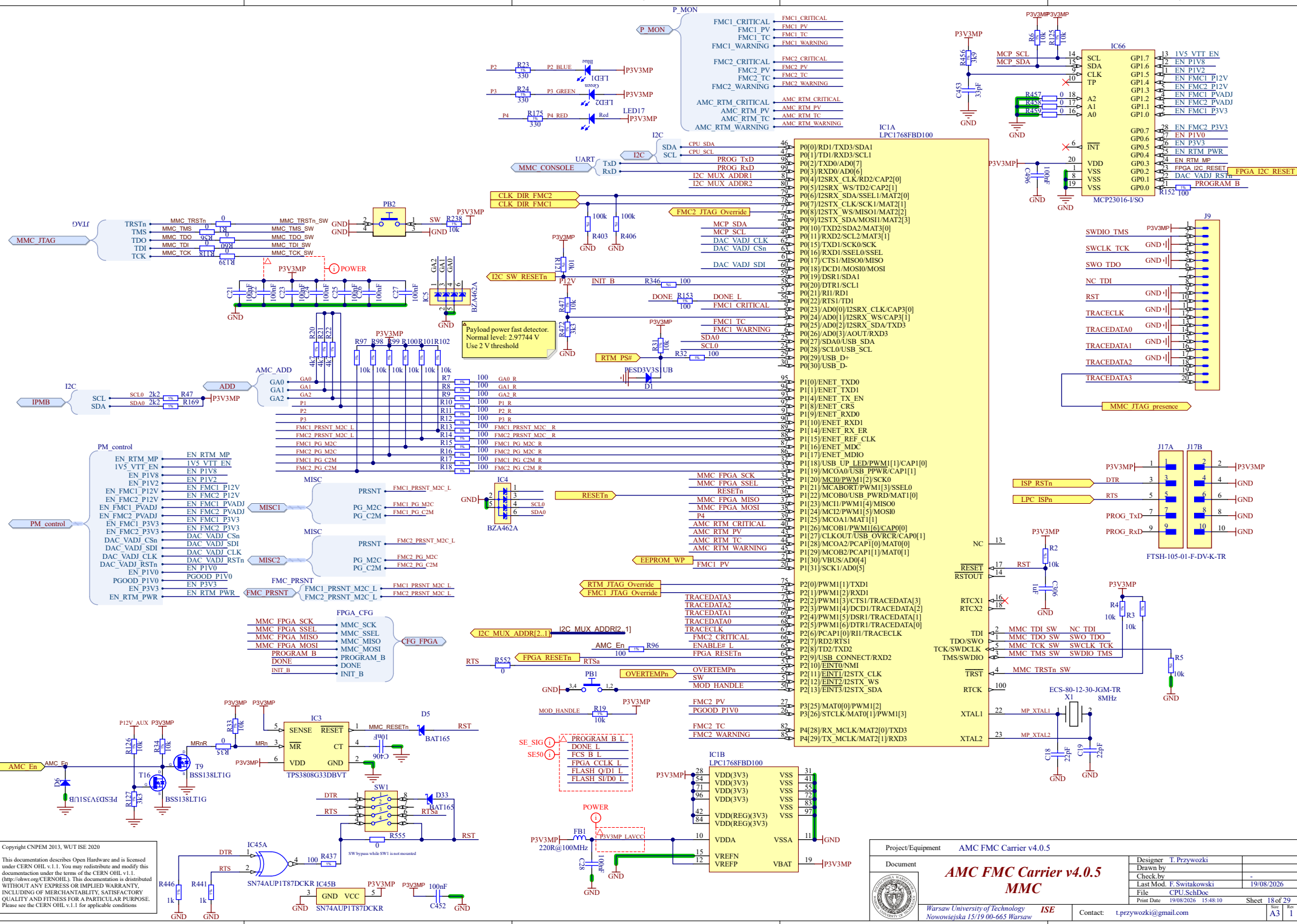
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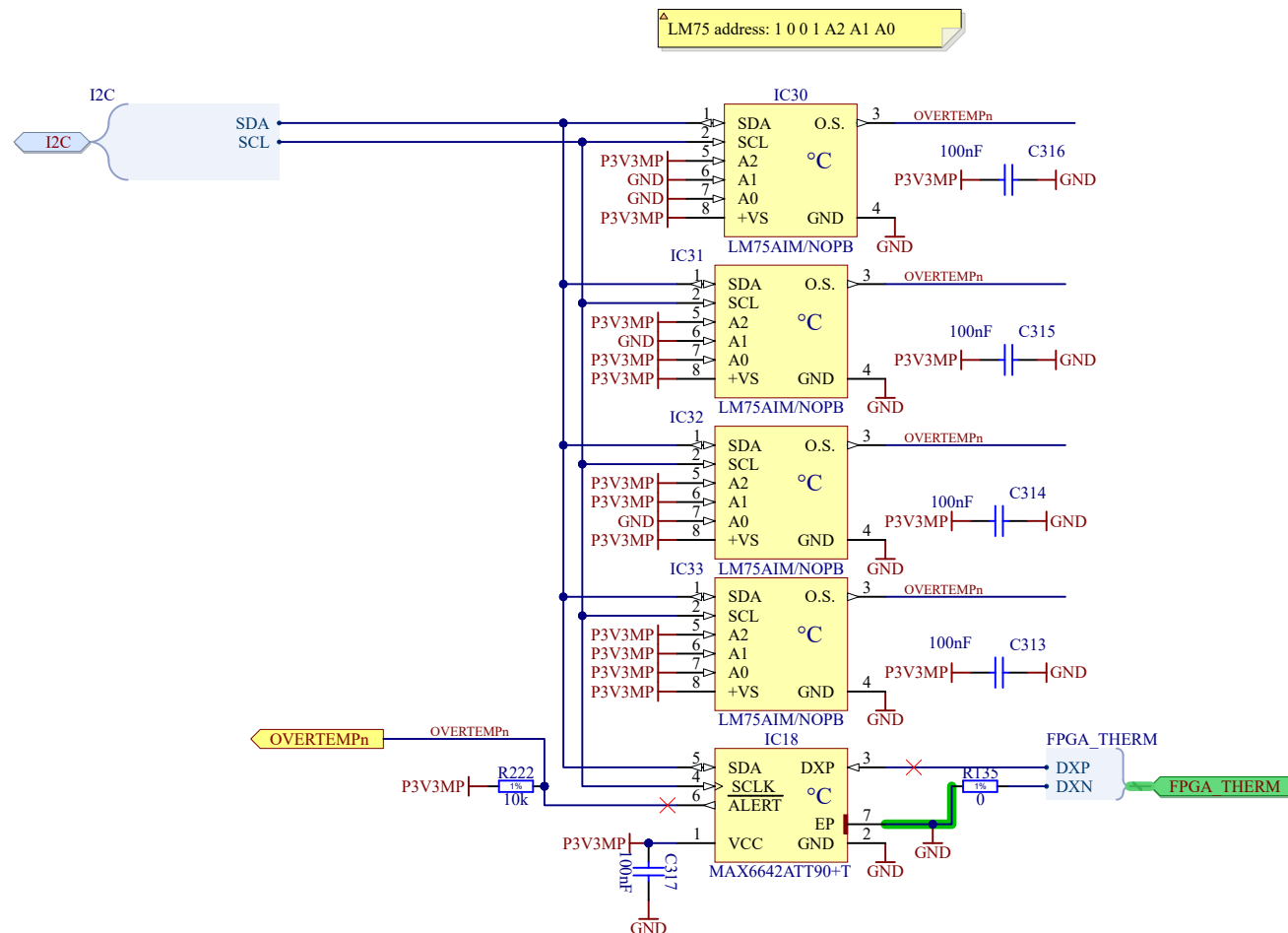


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Contact: t.przywozki@gmail.com

AMC FMC Carrier v4.0.5 DDR3 32bit





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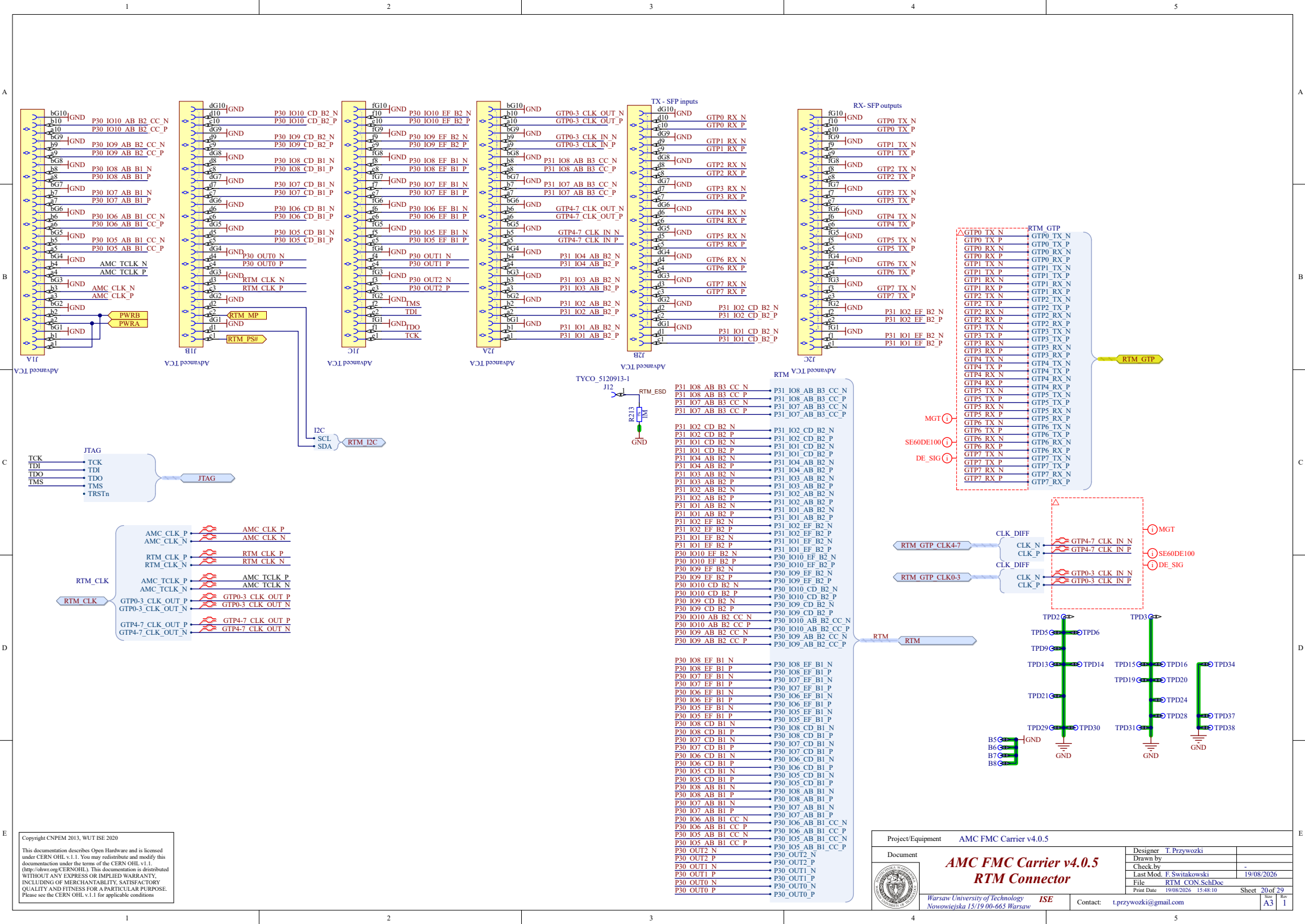
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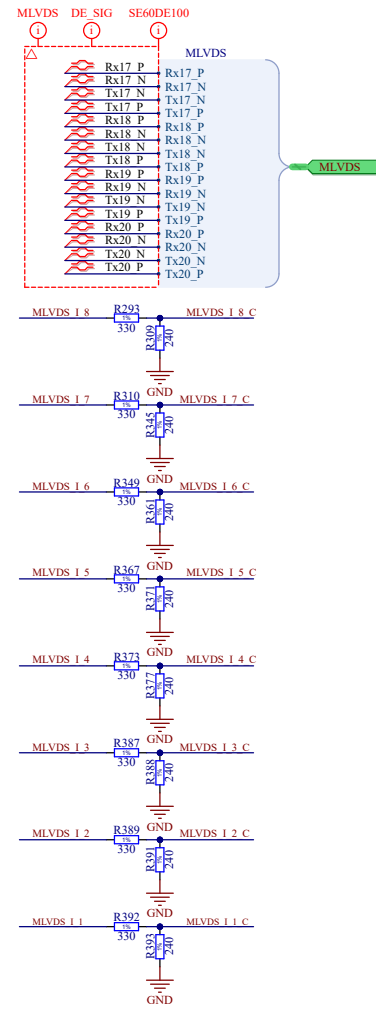
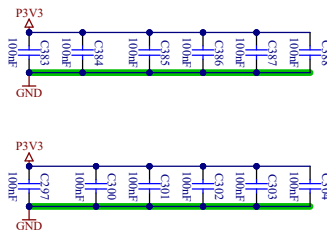
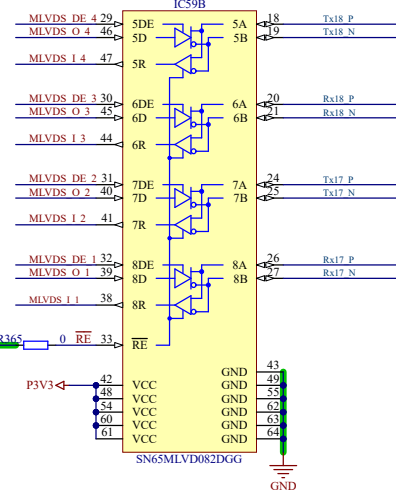
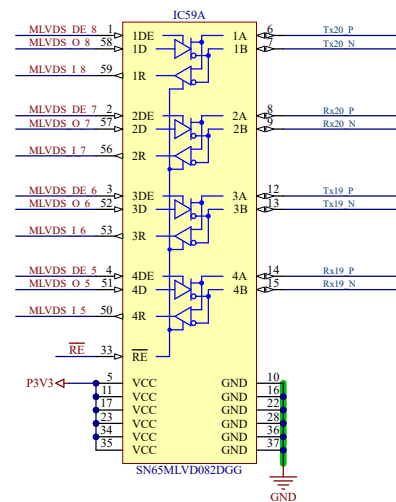
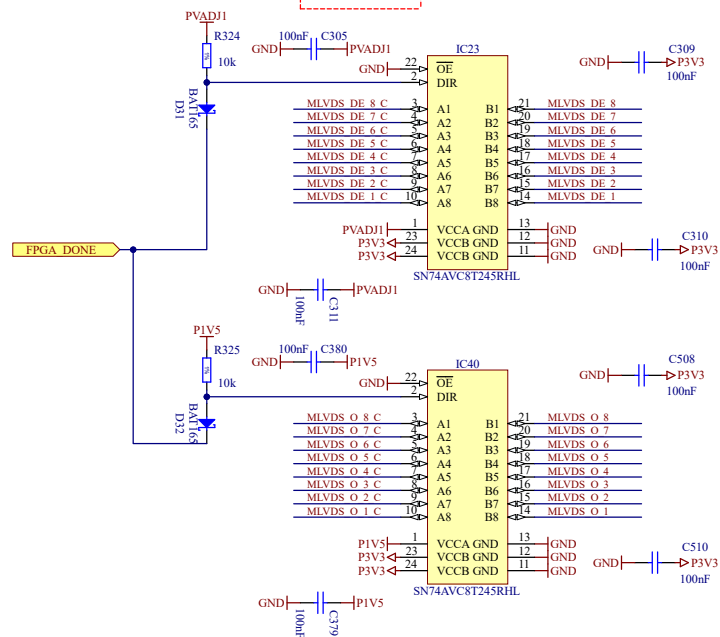
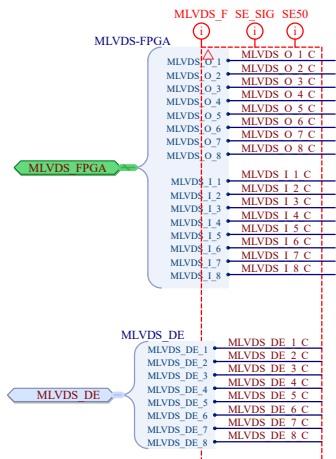
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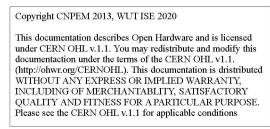
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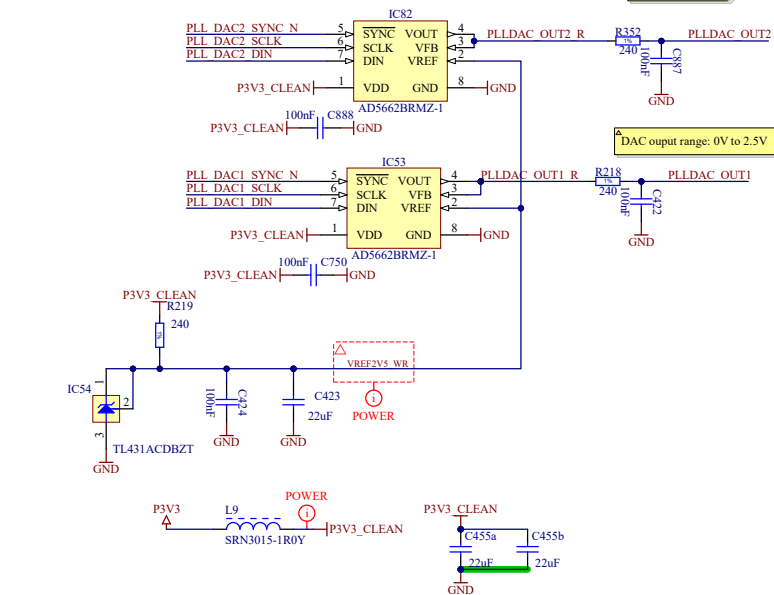
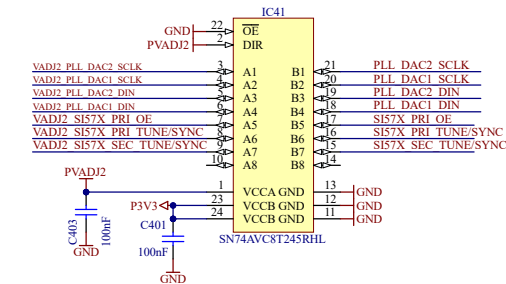
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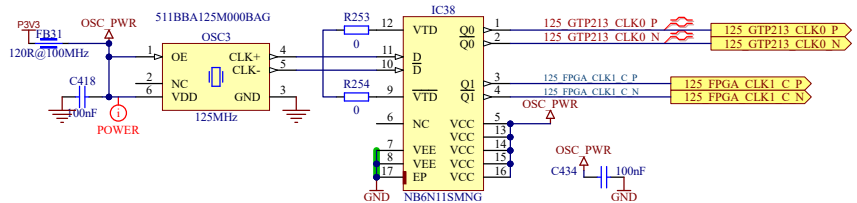
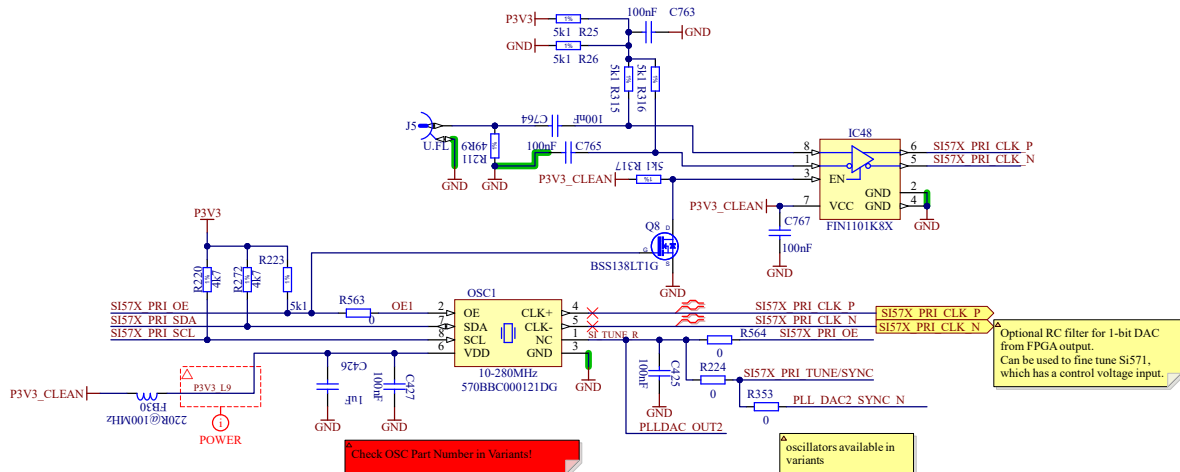
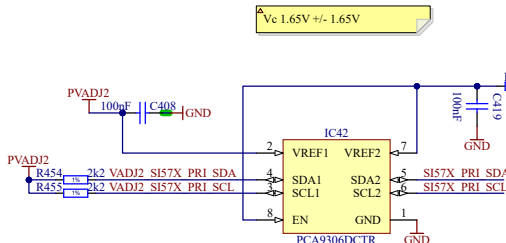
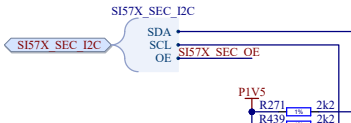
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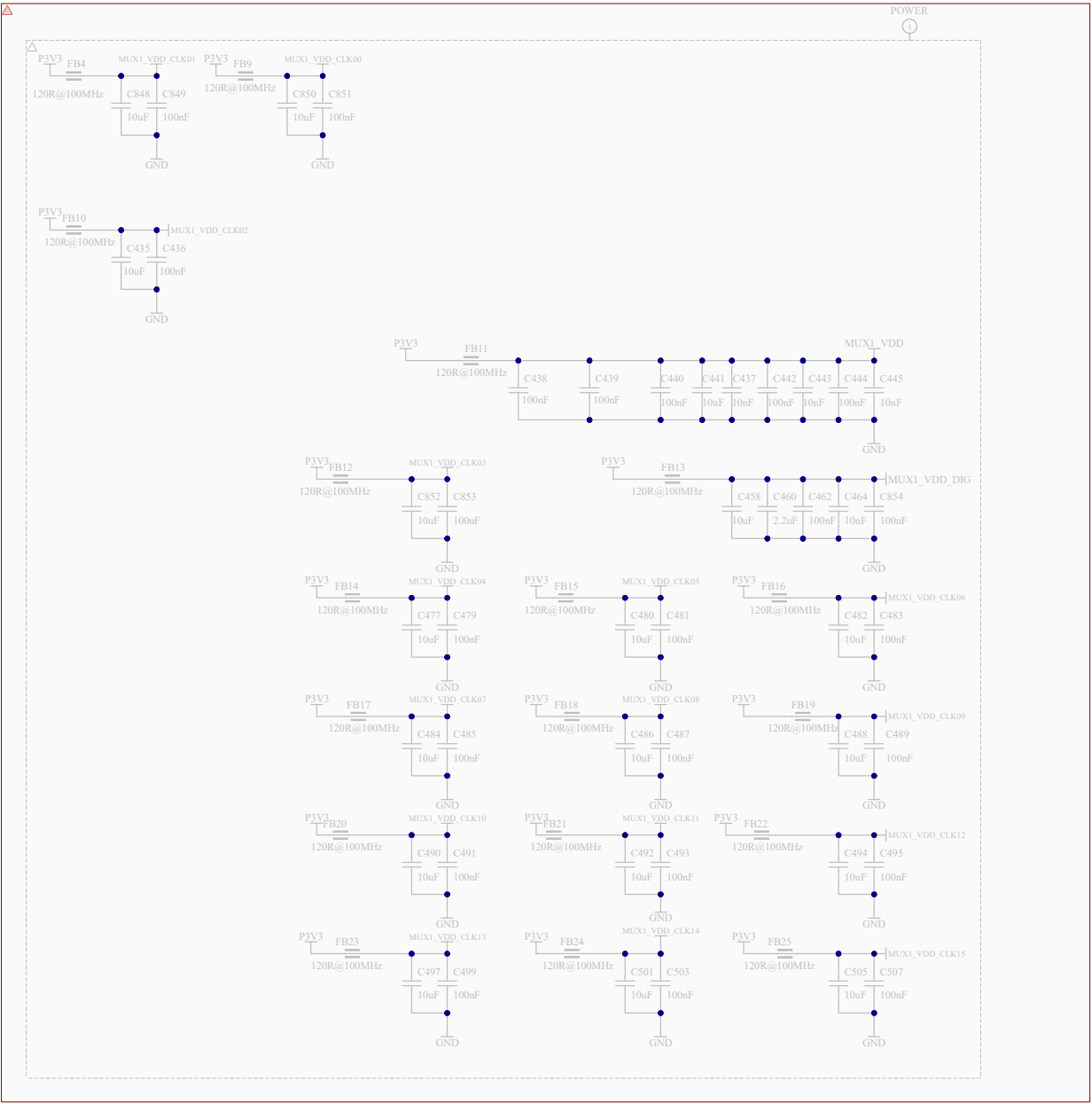




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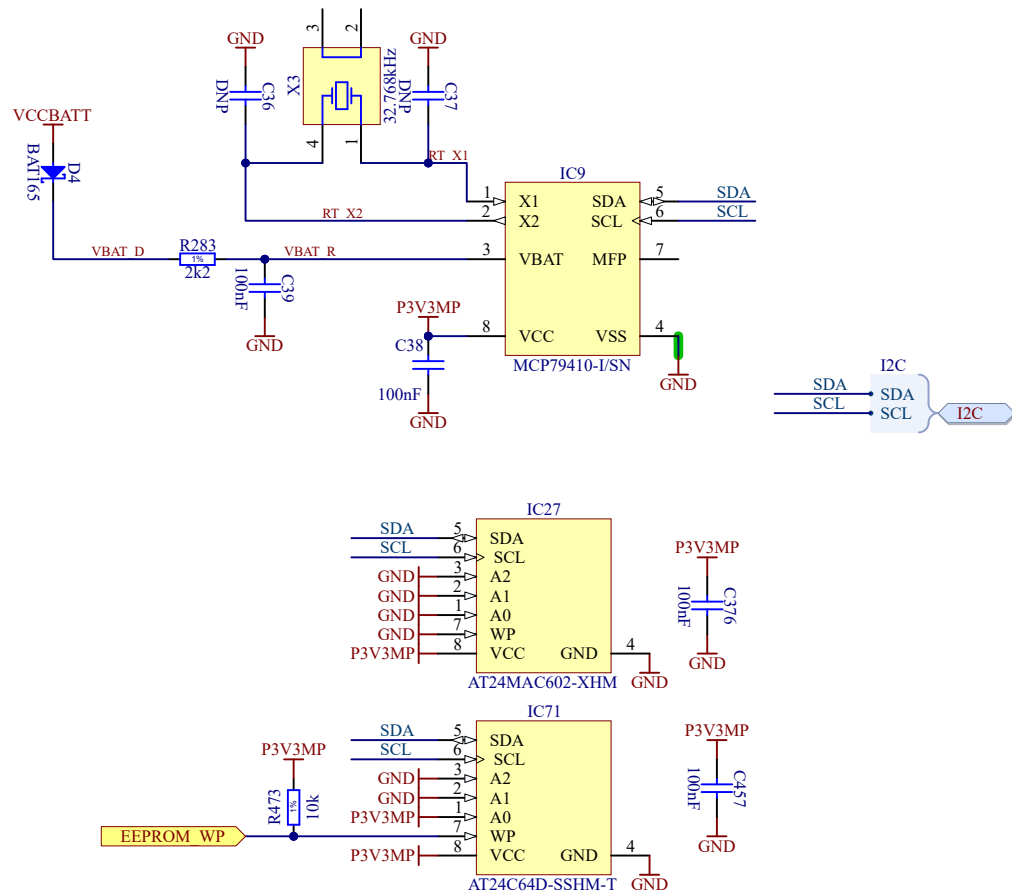




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Project/Equipment		AMC FMC Carrier v4.0.5	
Document		Designer T. Przywozki	
		Drawn by	
		Check by	
		Last Mod. F. Switakowski	
		File 8V54816A_SUPPLY.SchDoc	
		Print Date 19/08/2026 15:48:11	
Warsaw University of Technology Nowowiejska 15/19 00-663 Warsaw		Contact: t.przywozki@gmail.com	
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A3		Rev 1	



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Project/Equipment	AMC FMC Carrier v4.0.5		
Document	AMC FMC Carrier v4.0.5 RTCE & EEPROM		
Designer	T. Przywozki	Drawn by	XX/XX/XXXX
Check by	-	Last Mod.	F. Świtakowski
File	RTCE.SchDoc	Print Date	19/08/2026 15:48:11
Sheet	26 of 29	Size	A4
Contact:	t.przywozki@gmail.com		
		Rev	1

A

B

C

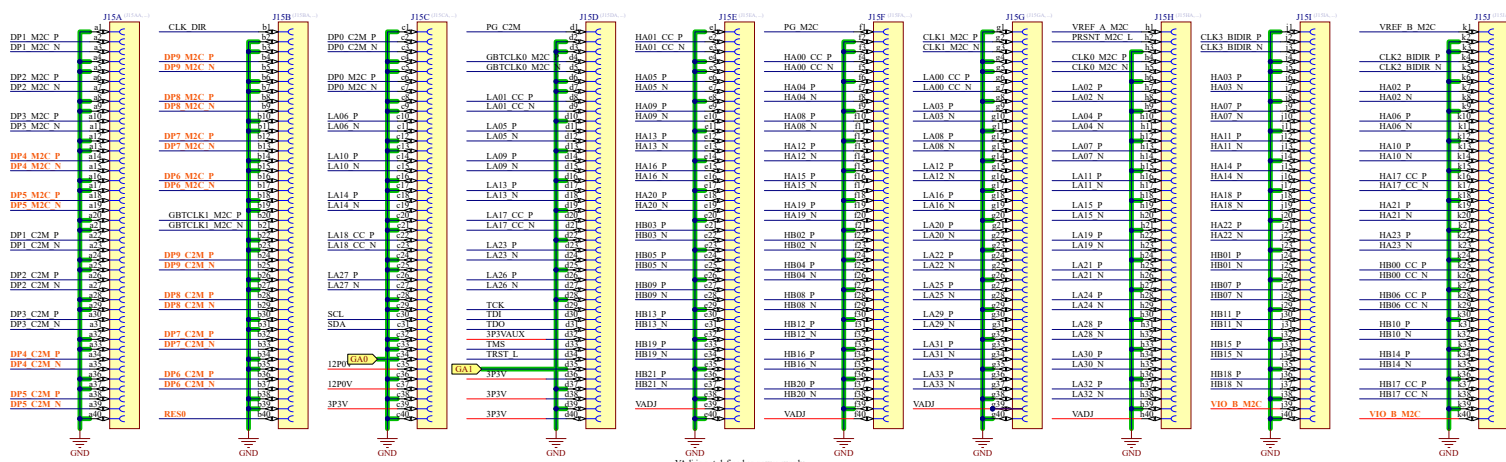
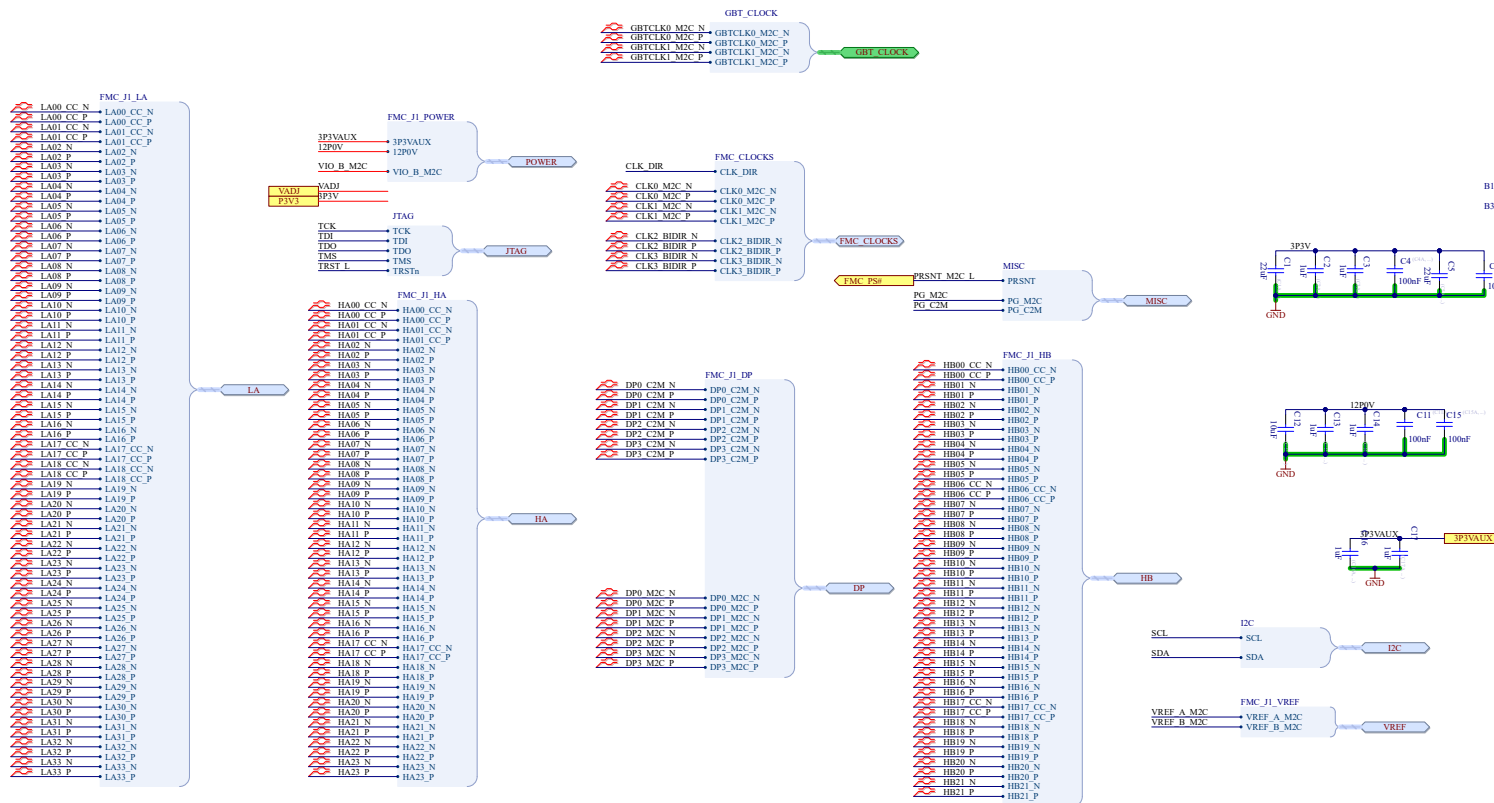
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A

B

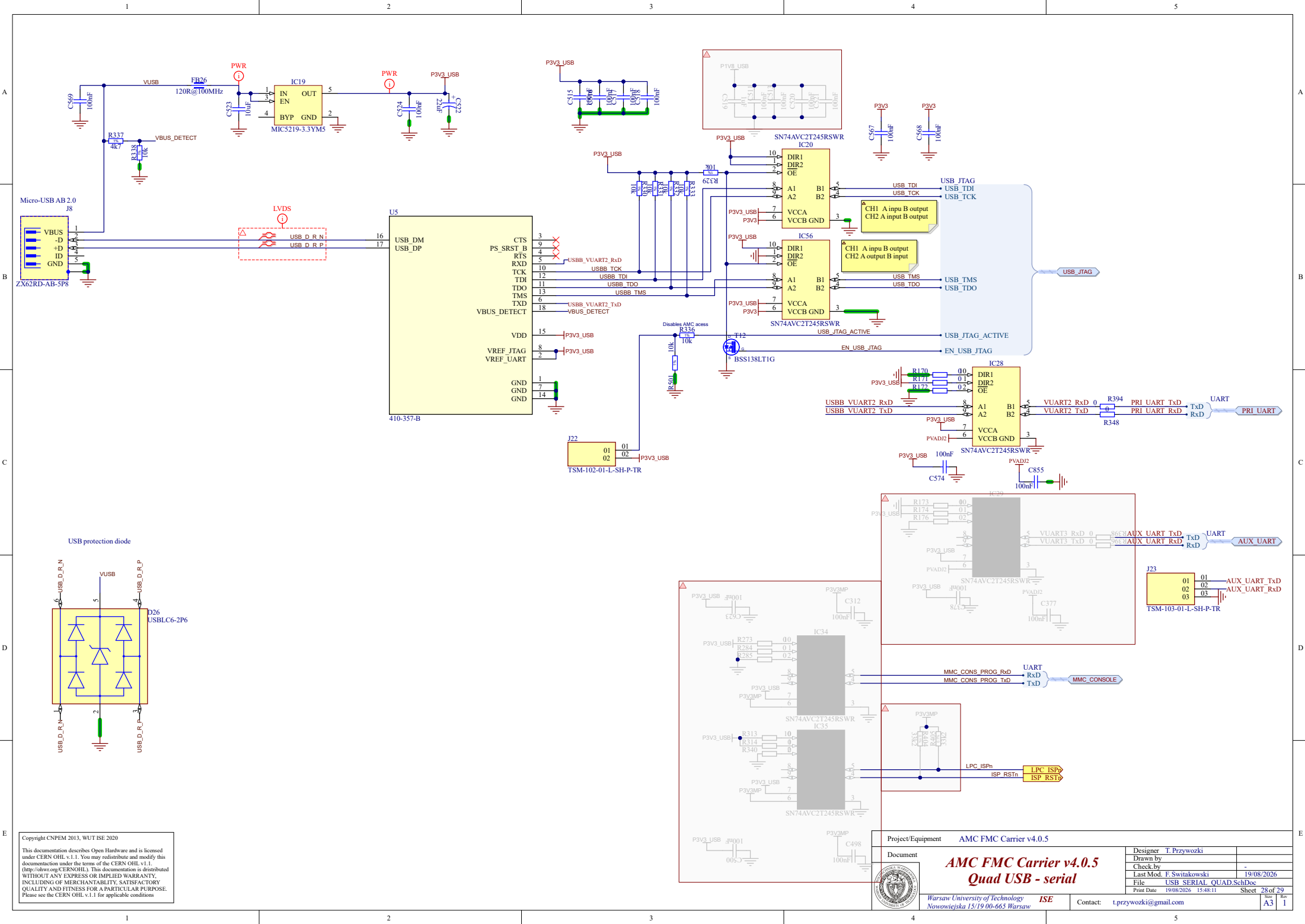
C

D



Vadj is not defined as power supply and therefore as global signal as it might differ for the 2 FMC

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Project/Equipment		AMC FMC Carrier v4.0.5	
Document		Designer T. Przywozki	
		Drawn by	
		Check by	
		Last Mod. F. Switkowski	
		19/08/2026	
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		Sheet 28 of 29	
		Rev A3	
		t.przywozki@gmail.com	

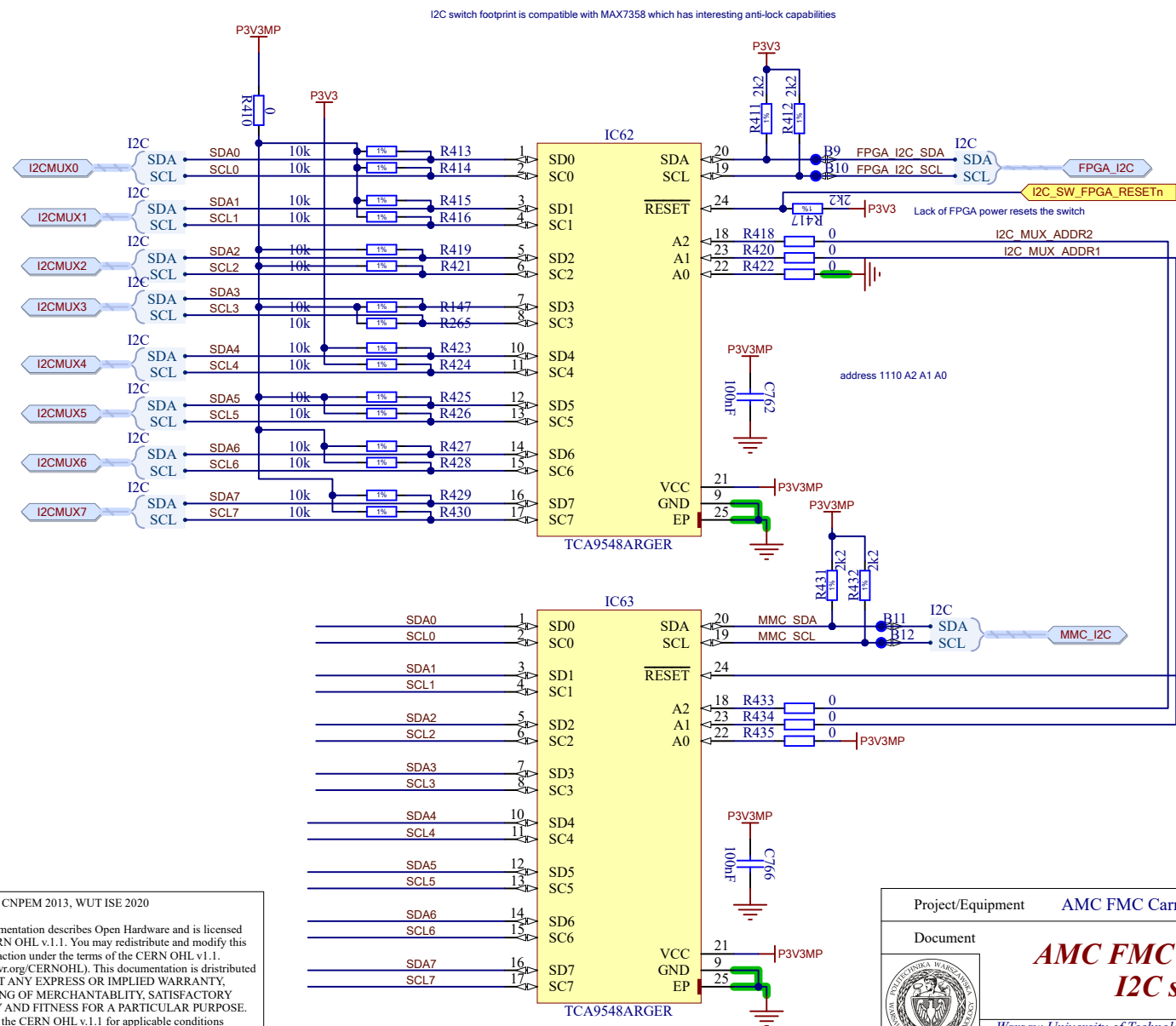


AMC FMC Carrier v4.0.5
Quad USB - serial

Warsaw University of Technology ISE
Nowowiejska 15/19 00-665 Warsaw

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I2C device address map

The resources are available from TCA9548s SW:
 MMC I2C1, addr 1110 A2 A1 0
 FPGA I2C and MMC I2C2, addr 1110 A2 A1 1
 where A2=P2[28], A1=P2[29]
 default addr is 0x70 and 0x71

MUX Port 0	MAX6642 0x48
Temp sensors	LM75: 0x4C LM75: 0x4D LM75: 0x4E LM75: 0x4F
MUX Port 1	RTC EEPROM: 0x57
RTCE	SRAM/RTCC reg: 0x6F EEPROM 2k: 0x50 EUI-ID: 0x59 EEPROM 64k: 0x51
MUX Port 2	
MUX Port 3	INA3221AIRGV: 0x40
Power	INA3221AIRGV: 0x41 INA3221AIRGV: 0x42
MUX Port 4	8V54816ANLG: 0x5B
Clock switch	
MUX Port 5	RTM devices
RTM	
MUX Port 6	EEPROM 0x52
FMC2	
MUX Port 7	EEPROM 0x50
FMC1	

▲ mux address 1110 A2 A1 0 from FPGA side 0x7x
 mux address 1110 A2 A1 1 from CPU side 0x7x

I2C_MUX_ADDR[2..1] I2C_MUX_ADDR[2..1]

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Project/Equipment AMC FMC Carrier v4.0.5

Document

AMC FMC Carrier v4.0.5 I2C switches

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